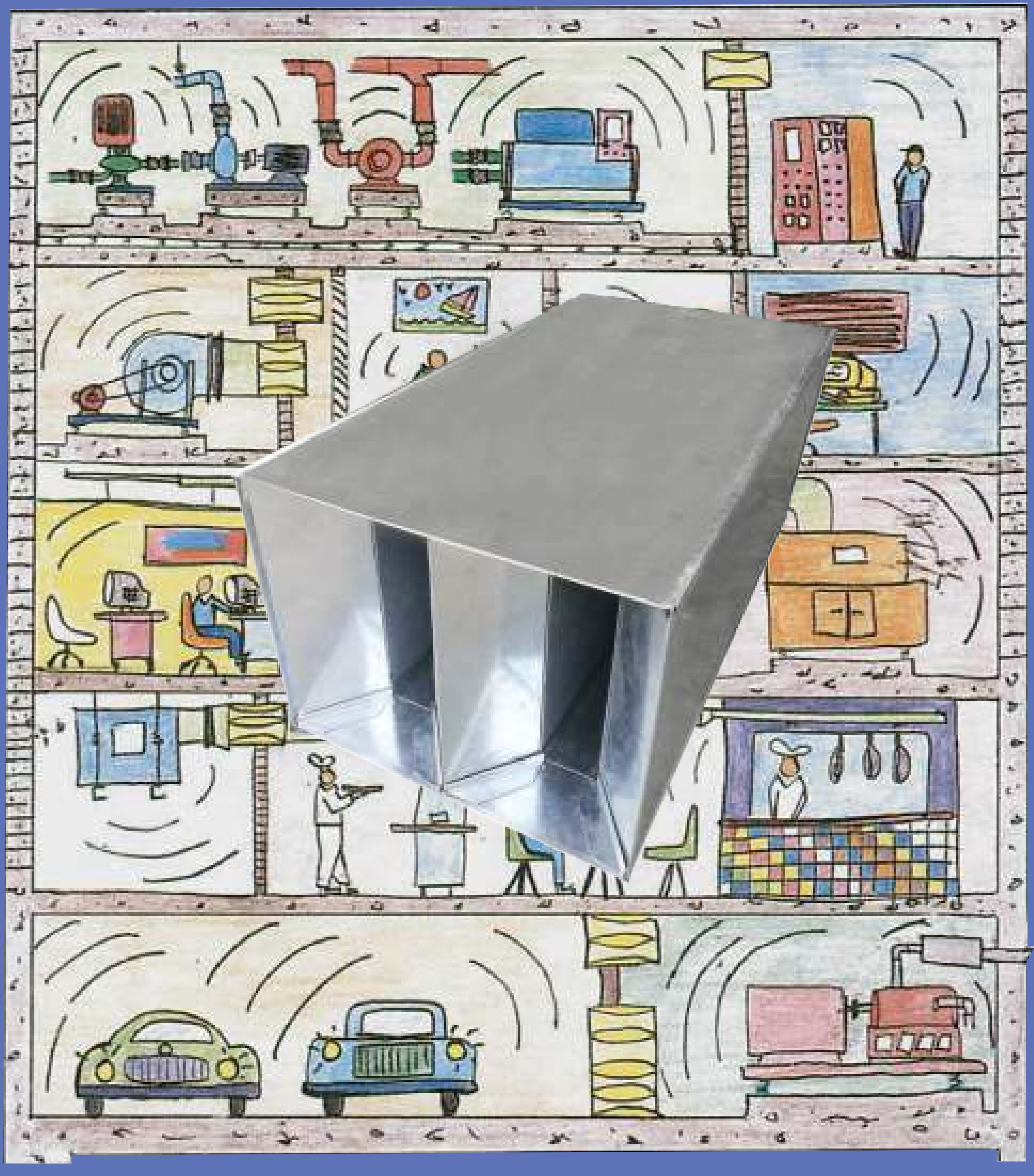
DUCT SILENCER

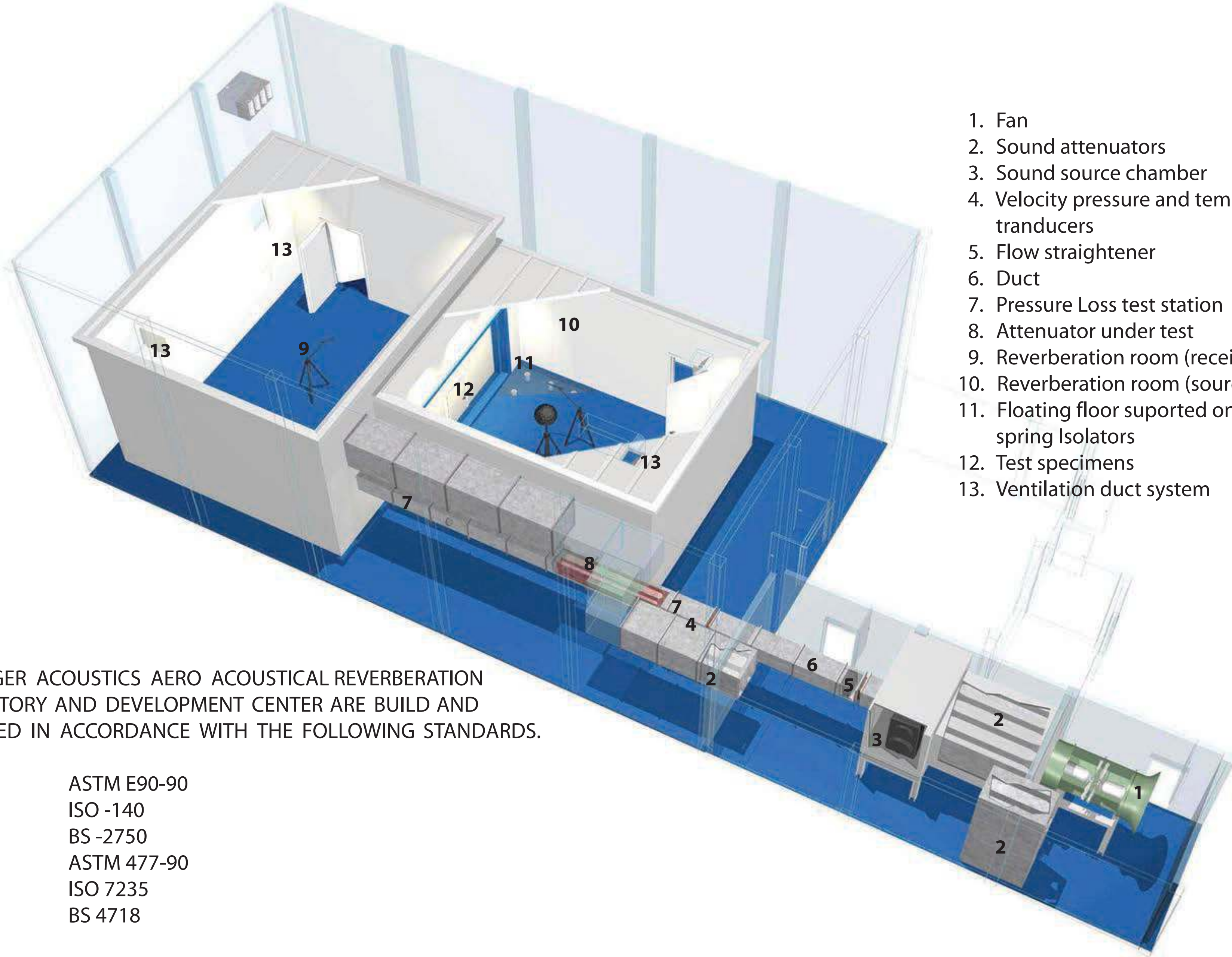
Abie Tiger



SOUND ATTENUATION
FOR A QUALITY OF LIFE

Publication No.501

ABIE TIGER ACOUSTICS AERO ACOUSTICAL REVERBERATION LABORATORY



1. Fan
2. Sound attenuators
3. Sound source chamber
4. Velocity pressure and temperature trandücers
5. Flow straightener
6. Duct
7. Pressure Loss test station
8. Attenuator under test
9. Reverberation room (receiving room)
10. Reverberation room (source room)
11. Floating floor suported on spring Isolators
12. Test specimens
13. Ventilation duct system

ABIE TIGER ACOUSTICS AERO ACOUSTICAL REVERBERATION LABORATORY AND DEVELOPMENT CENTER ARE BUILD AND OPERATED IN ACCORDANCE WITH THE FOLLOWING STANDARDS.

ASTM E90-90
ISO -140
BS -2750
ASTM 477-90
ISO 7235
BS 4718


System fan

System fan

1. The air flow system is powered by an axial flow fan capable of forward and reverse air flows to 42,475 m³/hr againist astatic pressure of 1494 pa.


Sound Attenuators

Sound Attenuators

2. The duct which runs from the fan room to the receiving room contains a sound source chamber. The attenuators under test are mounted within the duct between the sound source chamber and the receiving room.


Sound source chamber

Sound source chamber

3. The sound source chamber is constructed from 1.6 mm galvanized steel with an inner lining of fiberglass and perorated 0.8 mm galvanized steel. The sound source is a Brüel & kjaer calibrated sound source.


The duct system

The duct system

6. The duct system between the source chamber and receiving room is 610 x 610 mm. It is constructed from 1.6 mm galvanized steel as a sandwich construction with inner and outer skins with special infil and damping materials. The duct system is supported on 50 mm static deflection spring isolators.


Receiving room

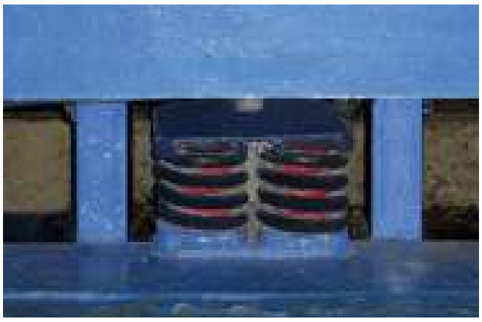
Receiving room

9. The receiving room is a reverberant space having a approximately volume of 300 m^3.

Source room

Source room

10. The source room is a reverberant space having a approximately volume of 200 m³.



Floating floor

Floating floors

11. Both receiving (9) and source (10) rooms are located floating floors on 25 and 50 mm deflection spring isolators. These assist in providing the acoustic isolation required between the two rooms.

Test specimen

Test specimen

12. Test specimens are mounted into an isolated frame seperating the receiving (9) and source rooms (10). The frame is supported on bridge bearing isolators with a minimum natural frequency of 10 Hz. Test specimens up to 3,010 mm high x 4,020 mm wide can fit into this frame.

Control room

Control room

14. The facility has a computer controlled air conditioned control room to remotely manage all aspect of the acoustic tests. In the control room, reverberation rooms, and source chamber are a wide range of state -of-the-art Bruel & Kjaer sound source and measuring equipment to meet the high standards required from this test facility.

FIRE TEST FOR ABIE TIGER SOUND LOCK SILENCERS

Sound attenuators which are placed in ducts reduce noise which is generated by airflow and by machinery connected to the duct. It is critical that in fire conditions, attenuators do not collapse and block airflow as the duct may be required to deliver clean air to people remaining in the building or to exhaust smoke from a fire affected area.

After completion of product development which involved finite element analysis to predict the mechanical properties of the attenuators, Abie Tiger submitted a test samples of various types of silencers including large unit models, for fire testing at Warrington Fire Research laboratory in Melbourne, Australia.

Two small unit attenuators and a large unit splitter were tested simultaneously accordance with BS 476 part 24-1987. Both were rated for 61 minutes stability and 61 minutes integrity.

After completion of the test, the splitters were examined for sustained local damage. All of the units tested were in good condition and free from cracking, distortion or collapse.

The only evidence that they had been subjected to fire test was some discolouration of the steel sheet coating.

DUCT SILENCER CONNECTIONS

METHODS OF ASSEMBLE

Multiple modules can be assembled into one of duct silencer module as detailed.

NOSE CONNECTION

Flanged cap : Fasten within 50 mm of corners at 200 mm intervals.

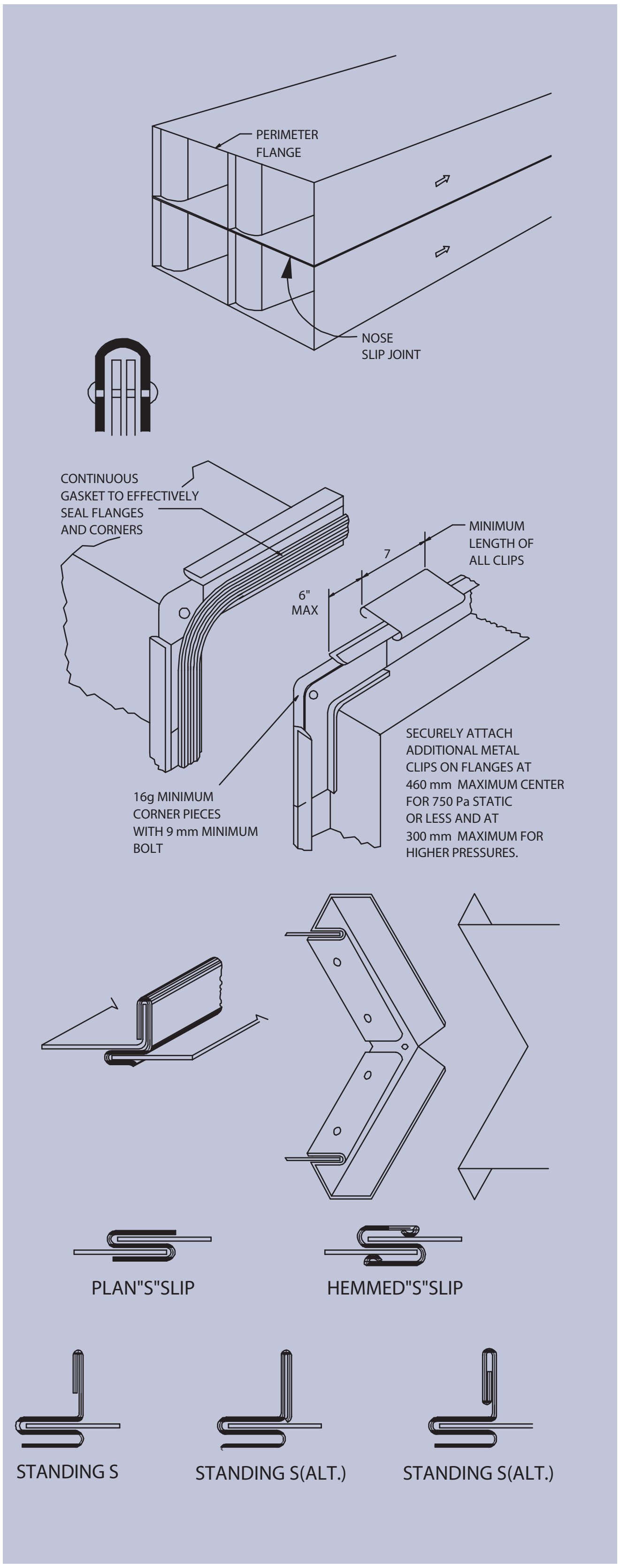
PERIMETER CONNECTION

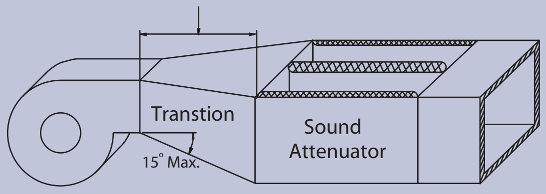
Formed Flange : Mating flanges are formed on the ends of the duct in double flange style to create a tee shape when assemblied. Minimum 16 guage steel corner pieces with 9 mm minimum diameter bolts are used toclose corners. 6 mm by 12 mm Minimum gaskets of suitable density and resiliency provied a continuous joint and are located to form an effective seal. Mating flanges are secured with 150 mm long clips located within 150 mm of each corner and also speced at centers not exceeding 375 mm for 750 Pa or less static pressure, nor exceeding 300 mm for 1000, 1500, 2500 Pa.

Pocket Lock : Can be used on all four sides of the duct. A 20 mm flange is used for 25 mm high lock, 30 mm flange for 35 mm lock. A button punch is used within 50 mm of corner and at 150 mm intervals, 750 pa maximum.

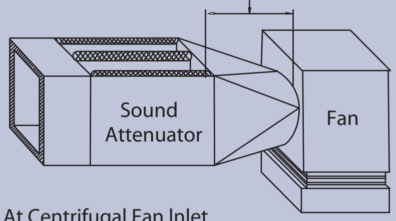
Flat S slips : Gauge shall be 24 minumum but not less than two gauges thinner than duct. When used on all four sides fasten within 50 mm of the corners and at 300 mm maximum intervals, 500 Pa maximum staticpressure.

Standing S : Slips can be used on all four sides to fasten the attenuator sections to the duct within 50 mm of the corner and at 300 mm maximum intervals. A 100 mm maximum width is used.

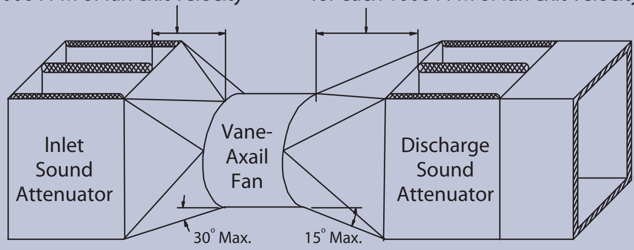




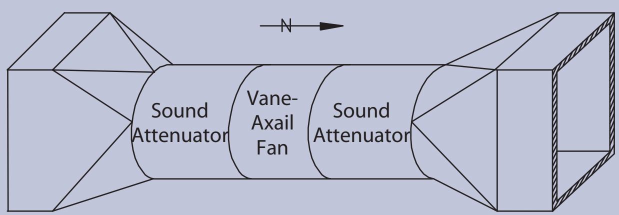
Centrifugal Fan Discharge
Note : Attenuator baffles should be perpendicular to fan shaft.



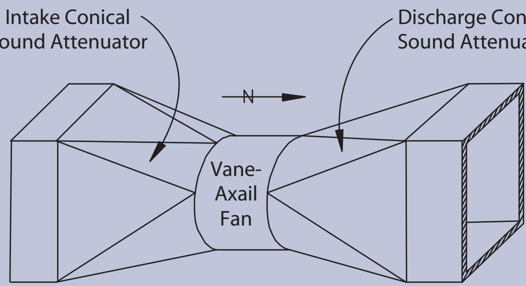
At Centrifugal Fan Inlet



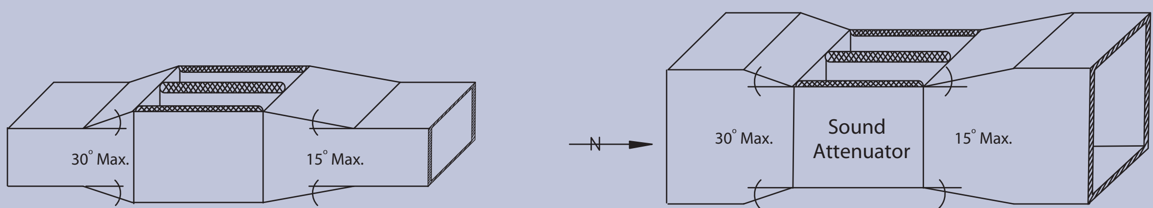
Rectangular Attenuator at Vane-Axial Fan
Inlet and Discharge



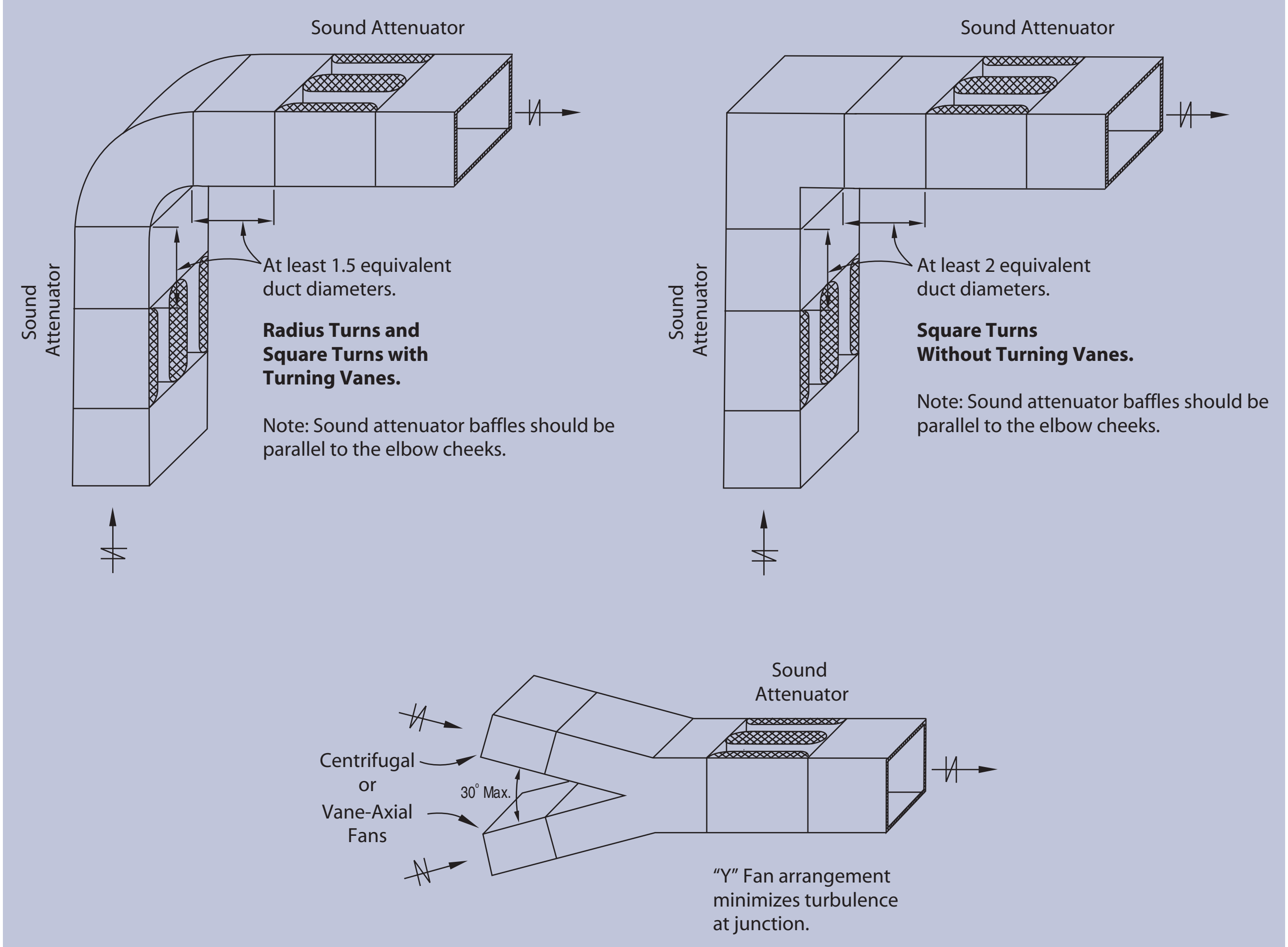
Circular Attenuators Attached
Directly To Vane-Axial Fan

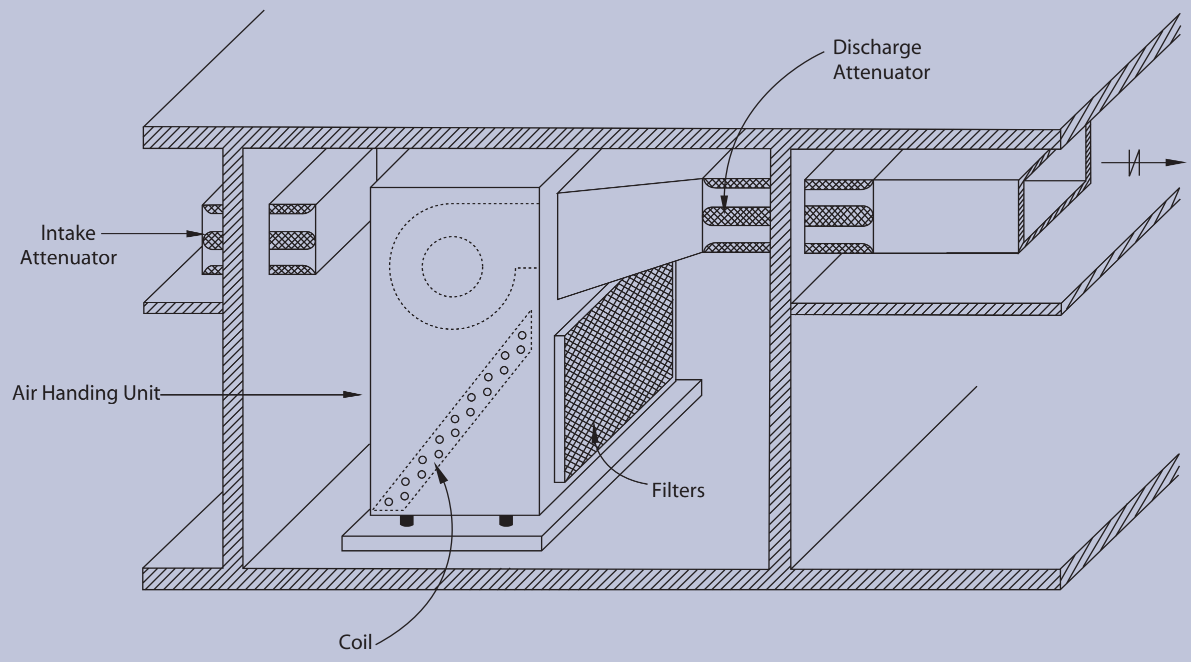


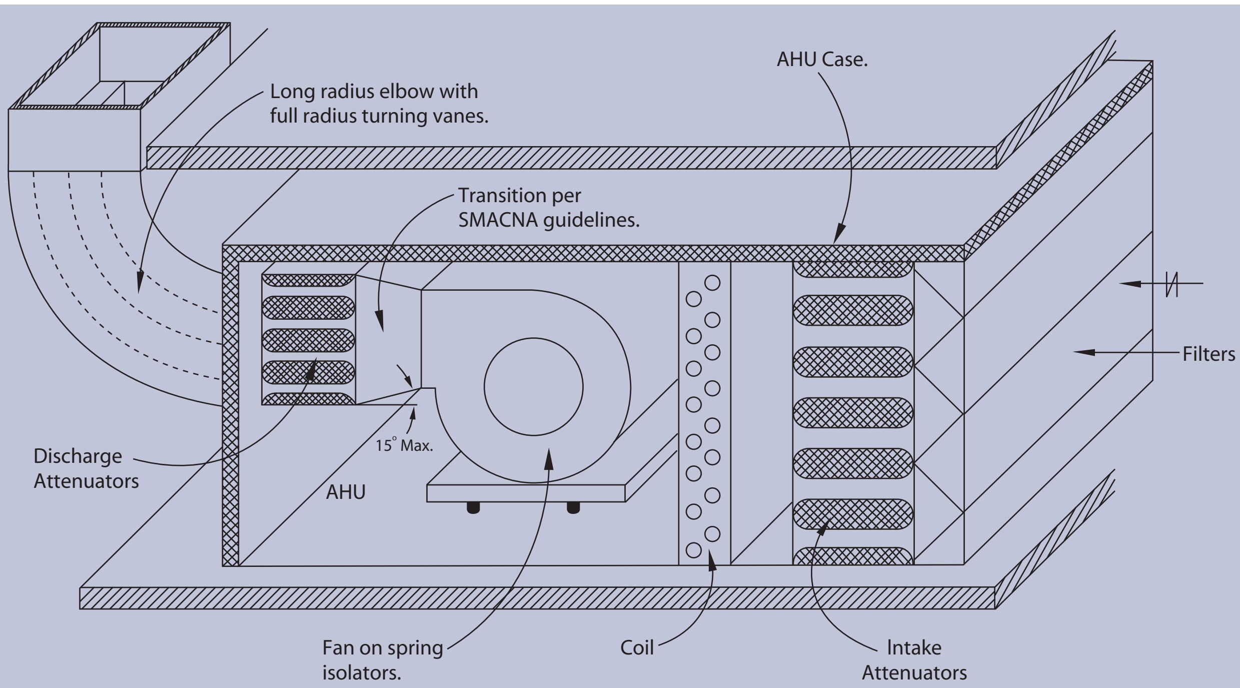
Circular Attenuators Attached
Directly To Vane-Axial Fan

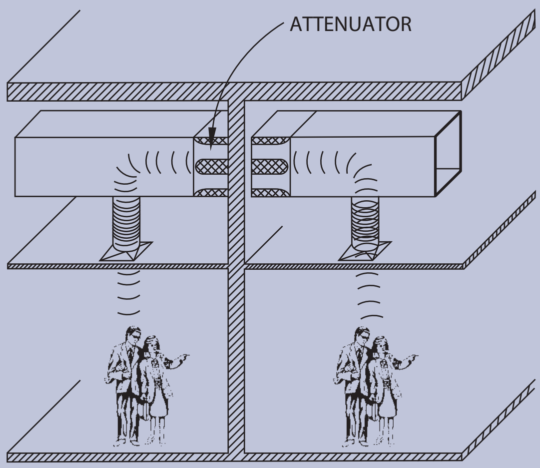


Upstream and Downstream of Transition

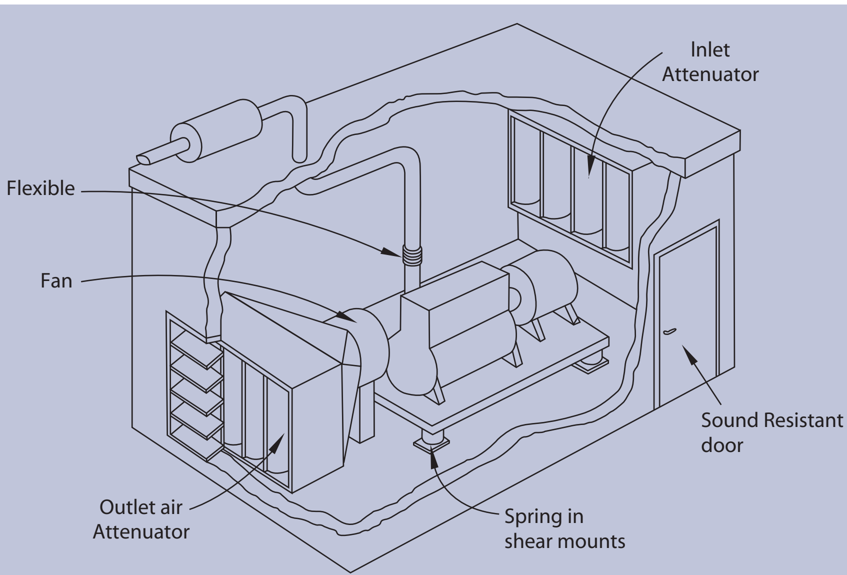

Sound Attenuator
Sound Attenuator
At least 1.5 equivalent duct diameters.
Radius Turns and Square Turns with Turning Vanes.
Note: Sound attenuator baffles should be parallel to the elbow cheeks.
Sound Attenuator
Sound Attenuator
At least 2 equivalent duct diameters.
Square Turns Without Turning Vanes.
Note: Sound attenuator baffles should be parallel to the elbow cheeks.
Sound Attenuator
Centrifugal or Vane-Axial Fans
30° Max.
"Y" Fan arrangement minimizes turbulence at junction.


Discharge
Attenuator
Intake
Attenuator
Air Handing Unit
Filters
Coil


Long radius elbow with
full radius turning vanes.
AHU Case.
Transition per
SMACNA guidelines.
Filters
15° Max.
Discharge
Attenuators
AHU
Fan on spring
isolators.
Coil
Intake
Attenuators



Insert duct attenuator into the duct
between rooms to reduce cross-talk noise.



Diagramatic view of generator house.

REFERENCE SOURCE :

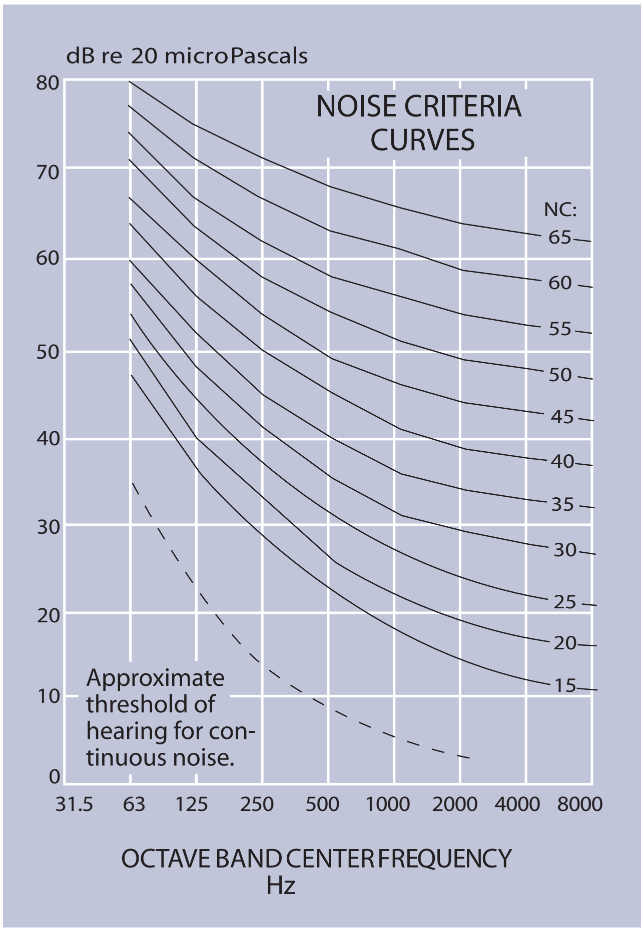
- American Society of Heading, Refrigerating, and Air-Conditioning Engineers, inc., Attanta. ASI Practical Guide to Noise and Viberation Control for HVAC System.

CRITERIA FOR DESIGN
NOISE CRITERIA CURVES (NC)

Recommended indoor Design Criteria for Air Conditioning System Sound Control[a]

Note : These are for unoccupied spaces with all systems operating

Type of area	Recommended NC Criteria range
Private residences	25 to 30
Apartments	25 to 30
Hotels/Motels	
Individual rooms or suites	30 to 35
Meeting/banquet rooms	25 to 30
Halls, corridors, lobbies	35 to 40
Service/support areas	40 to 45
Offices	
Executive	25 to 30
Conference rooms	25 to 30
Private	30 to 35
Open-plan areas	35 to 40
Computer equipment rooms	40 to 45
Public circulation	40 to 45
Hospitals and clinics	
Private rooms	25 to 30
Wards	30 to 35
Operating rooms	35 to 40
Corridors	35 to 40
Public areas	35 to 40
Churches	25 to 30 b
Schools	
Lecture theater and classrooms	30 to 35
Open-plan classrooms	30 to 35
Libraries	35 to 40
Concert halls	b
Recording studios	b
Movie theaters	30 to 35



Source : ASHRAE Handbook, 1987 Systems. Chapter 52,Sound 3 Vibration Control, American Society of Heating, Refrigeration, and Air-Conditioning. a) Design goals can be increased by 5 dB when dictated by budget constraints or when noise intrusion from other sources represents a limiting condition. b) An acoustical expert should be consulted for guidance on these critical spaces.

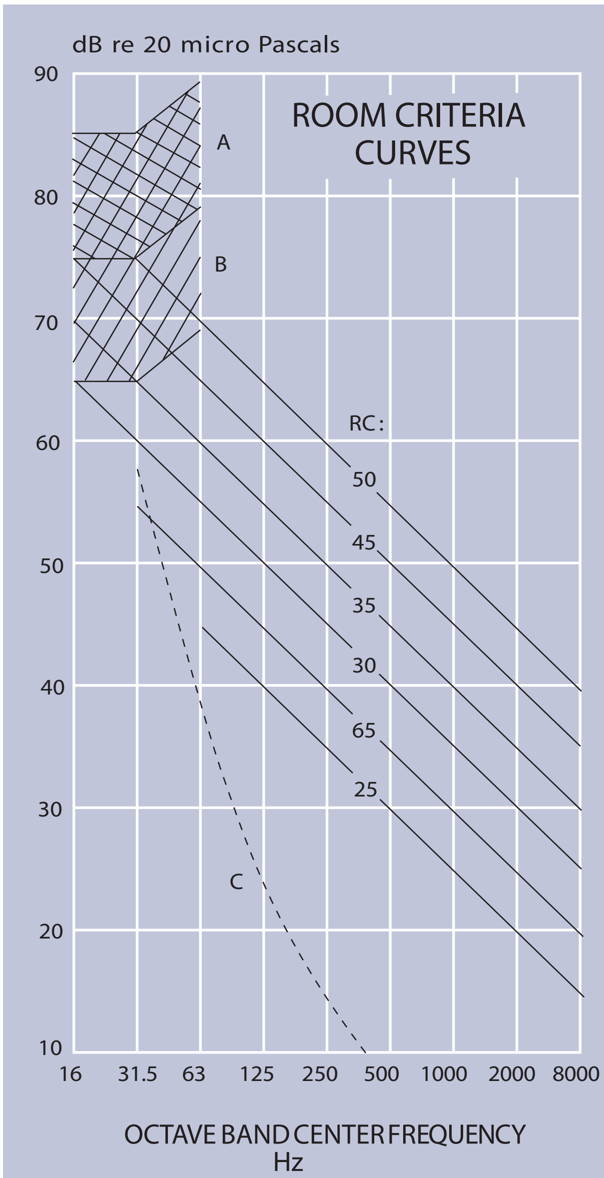
Nois Criteria	Octave Frequency Band In Hz							
Curve	**63**	**125**	**250**	**500**	**1000**	**2000**	**4000**	**8000**
NC-15	47	36	29	22	17	14	12	11
NC-20	51	40	33	26	22	19	17	16
NC-25	54	44	37	31	27	24	22	21
NC-30	57	48	41	35	31	29	28	27
NC-35	60	52	45	40	36	34	33	32
NC-40	64	56	50	45	41	39	38	37
NC-45	67	60	54	49	46	44	43	42
NC-50	71	64	58	54	51	49	48	47
NC-55	74	67	62	58	56	54	53	52
NC-60	77	71	67	63	61	59	58	57
NC-65	80	75	71	68	66	64	63	62

CRITERIA FOR DESIGN
ROOM CRITERIA CURVES (RC)

Recommended indoor Design Criteria for Air Conditioning System Sound Control[a]

Note : These are for unoccupied spaces with all systems operating

Type of area	Recommended RC Criteria range
Private residences	25 to 30
Apartments	25 to 30
Hotels/Motels	
Individual rooms or suites	30 to 35
Meeting/banquet rooms	25 to 30
Halls, corridors, lobbies	35 to 40
Service/support areas	40 to 45
Officas	
Executive	25 to 30
Conference rooms	25 to 30
Private	30 to 35
Open-plan areas	35 to 40
Computer equipment rooms	40 to 45
Public circulation	40 to 45
Hospitals and Clinics	
Private rooms	25 to 30
Wards	30 to 35
Operating rooms	35 to 40
Corridors	35 to 40
Public areas	35 to 40
Churches	25 to 30 b
Schools	
Lecture and classrooms	30 to 35
Open-plan classrooms	30 to 35
Libraries	35 to 40
Concert halls	b
Recording studios	b
Movie theaters	30 to 35



Source : ASHRAE Handbook, 1987 Systems. Chapter 52, Sound 3 Vibration Control, American Society of Heating, Refrigeration, and Air-Conditioning.
a) Design goals can be increased by 5 dB when dictated by budget constraints or when noise intrusion from other sources represents a limiting condition.
b) An acoustical expert should be consulted for guidance on these critical spaces.

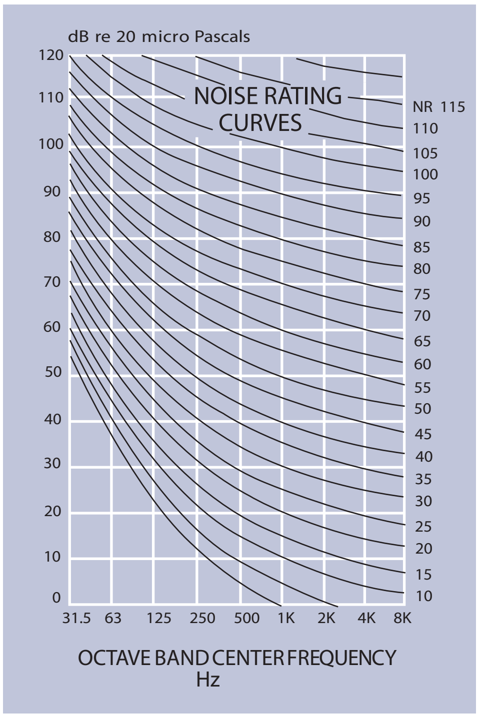
Region a. high probability that noise induced vibration levels in lightweight wall and celing construction will be clearly feelable anticipate audible rattles in light fixtures. Doors, window etc.
Region b. noise-induced vibration levels in lightweigt wall and ceiling constructions may be moderately feelable slight possibility of rattles in light fixtures, doors, windows. etc.
Region c. below threshold of hearing for continous noise
Source : ASHARE Handbook, 1987 Systems and Applications.

CRITERIA FOR DESIGN
NOISE CRITERIA CURVES (NR)

Recommended NR curves for various environments

Environment	NR Curve
Office Buildings	
General open offices, reception areas	NR 40
Conference rooms	NR 30
Executive offices	NR 35
Foyers, public areas	NR 45
Typing and machinist rooms, computer rooms	NR 45
Hospitals	
Hospital Wards	NR 35
Intensive care wards, operating theatres	NR 30
Laboratories, casualty areas	NR 40
Kitchens, sterilising and service areas	NR 45
Surgery, dental clinics, consulting rooms	NR 40
Waiting rooms and reception areas	NR 45
Schools	
Classrooms	NR 35
Lecture theatres, conference rooms and assembly halls	NR 30
Recreation halls, gymnasium	
Learning areas	NR 40
Workshops, laboratories	NR 45
Music practice rooms, office areas	NR 40
Radio and TV Studios	
Recording studios, news, rooms, interview rooms, etc	NR 25
Audience studios	NR 30
Auditoriums and Music Halls	
Concert and opera halls	NR 25
Music practice rooms	NR 30
Lecture halls, theatres	NR 30
Lobbies	NR 40
Hotels/Motels	
Dining rooms, restaurants	NR 40
Bedrooms, ballrooms, banquet rooms	NR 35
Kitchen, laundries, bars and lounges	NR 45
Public Buildings	
Administrative office space, reading areas	NR 35
Indoor Sports Buildings	
Billiards and snooker rooms	NR 45
Gymnasium, squash courts, bowling alleys	NR 50
Swimming Pools	NR 55
General Service Areas For All Buildings	
Corridors, toilets, washrooms,	NR 45
Plant rooms	NR 70

NR (ISO) Noise Curves - Developed by the international Standards Organization 1971 to rate noisiness with the 1000 Hz octave band as the reference point



Noise Rating Curves

Maximum Sound Pressure Level (dB)									
Noise Rating - NR - Curve	Octave band mid-frequency (Hz)								
	31.5	62.5	125	250	500	1000	2000	4000	8000
NR 0	55	36	22	12	5	0	-4	-6	-8
NR 10	62	43	31	21	15	10	7	4	2
NR 20	69	51	39	31	24	20	17	14	13
NR 30	76	59	48	40	34	30	27	25	23
NR 40	83	67	57	49	44	40	37	35	33
NR 50	89	75	66	59	54	50	47	45	44
NR 60	96	83	74	68	63	60	57	55	54
NR 70	103	91	83	77	73	70	68	66	64
NR 80	110	99	92	86	83	80	78	76	74
NR 90	117	107	100	96	93	90	88	86	85
NR 100	124	115	109	105	102	100	98	96	95
NR 110	130	122	118	114	112	110	108	107	105
NR 120	137	130	126	124	122	120	118	117	116
NR 130	144	138	135	133	131	130	128	127	126

REF : A

Estimated duct attenuation (in dB per 0.3 m) For bare, unlined straight metal ducts. "D" is the inside diameter of a round duct or smaller inside duct dimension of an oval or rectangular duct.

D,	Octave Frequency Band, Hz							
mm.	63	125	250	500	1000	2000	4000	8000
FOR RECTANGULAR (OR SQUARE) DUCTS :								
100 - 225	0.3	0.3	0.2	0.1	0.1	0.1	0.1	0.1
250 - 375	0.3	0.3	0.2	0.1	0.1	0.1	0.1	0.1
400 - 525	0.3	0.2	0.1	0.1	0.1	0.1	0.1	0.1
350 - 675	0.3	0.2	0.1	0.1	0.1	0.1	0.1	0.1
700 - 825	0.3	0.2	0.1	0.1	0.1	0.1	0.1	0.1
850 - 975	0.3	0.2	0.1	0.1	0.1	0.1	0.1	0.1
1000 - 1125	0.3	0.2	0.1	0.1	0.1	0.1	0.1	0.1
1150 - 1350	0.2	0.15	0.1	0.1	0.1	0.1	0.1	0.1
1375 - 1650	0.2	0.15	0.1	0.1	0.1	0.1	0.1	0.1
1675 - 1950	0.2	0.15	0.1	0.1	0.1	0.1	0.1	0.1
1975 - 2250	0.2	0.15	0.1	0.1	0.1	0.1	0.1	0.1
> OVER 2250	0.2	0.15	0.1	0.1	0.1	0.1	0.1	0.1
FOR OVAL OR ROUND DUCTS :								
100 - 225	0.15	0.15	0.1	0.05	0.05	0.05	0.05	0.05
250 - 375	0.15	0.15	0.1	0.05	0.05	0.05	0.05	0.05
400 - 525	0.15	0.1	0.05	0.05	0.05	0.05	0.05	0.05
350 - 675	0.15	0.1	0.05	0.05	0.05	0.05	0.05	0.05
700 - 825	0.15	0.1	0.05	0.05	0.05	0.05	0.05	0.05
850 - 975	0.15	0.1	0.05	0.05	0.05	0.05	0.05	0.05
1000 - 1125	0.15	0.1	0.05	0.05	0.05	0.05	0.05	0.05
1150 - 1350	0.1	0.08	0.05	0.05	0.05	0.05	0.05	0.05
1375 - 1650	0.1	0.08	0.05	0.05	0.05	0.05	0.05	0.05
1675 - 1950	0.1	0.08	0.05	0.05	0.05	0.05	0.05	0.05
1975 - 2250	0.1	0.08	0.05	0.05	0.05	0.05	0.05	0.05
>OVER 2250	0.1	0.08	0.05	0.05	0.05	0.05	0.05	0.05

REF : B

Estimated duct attenuation (in dB per 0.3 m) For Straight metal ducts with an outsidewrapping of 25 mm fiberglass or some thick thermal insulationmaterial. "D" is theinside diameter of a round duct or smallerinside duct with dimension of oval or rectangular duct.

D,	Octave Frequency Band, Hz							
mm.	63	125	250	500	1000	2000	4000	8000
FOR RECTANGULAR (OR SQUARE) DUCTS :								
100 - 225	0.4	0.4	0.25	0.15	0.1	0.1	0.1	0.1
250 - 375	0.4	0.4	0.25	0.15	0.1	0.1	0.1	0.1
400 - 525	0.4	0.3	0.15	0.15	0.1	0.1	0.1	0.1
350 - 625	0.4	0.3	0.15	0.15	0.1	0.1	0.1	0.1
700 - 825	0.4	0.3	0.15	0.15	0.1	0.1	0.1	0.1
850 - 975	0.4	0.3	0.15	0.15	0.1	0.1	0.1	0.1
1000 - 1125	0.4	0.3	0.15	0.15	0.1	0.1	0.1	0.1
1150 - 1350	0.3	0.25	0.15	0.15	0.1	0.1	0.1	0.1
1375 - 1650	0.3	0.25	0.15	0.15	0.1	0.1	0.1	0.1
1675 - 1950	0.3	0.25	0.15	0.15	0.1	0.1	0.1	0.1
1975 - 2250	0.3	0.25	0.15	0.15	0.1	0.1	0.1	0.1
> OVER 2250	0.3	0.25	0.15	0.15	0.1	0.1	0.1	0.1
FOR OVAL OR ROUND DUCTS :								
100 - 225	0.25	0.25	0.15	0.08	0.05	0.05	0.05	0.05
250 - 375	0.25	0.25	0.16	0.08	0.05	0.05	0.05	0.05
400 - 525	0.25	0.2	0.1	0.08	0.05	0.05	0.05	0.05
350 - 625	0.25	0.2	0.1	0.08	0.05	0.05	0.05	0.05
700 - 825	0.25	0.2	0.1	0.08	0.05	0.05	0.05	0.05
850 - 975	0.25	0.2	0.1	0.08	0.05	0.05	0.05	0.05
1000 - 1125	0.25	0.2	0.1	0.08	0.05	0.05	0.05	0.05
1150 - 1350	0.2	0.15	0.08	0.06	0.05	0.05	0.05	0.05
1375 - 1650	0.2	0.15	0.08	0.06	0.05	0.05	0.05	0.05
1675 - 1950	0.2	0.15	0.08	0.06	0.05	0.05	0.05	0.05
1975 - 2250	0.2	0.15	0.08	0.06	0.05	0.05	0.05	0.05
>OVER 2250	0.2	0.15	0.08	0.06	0.05	0.05	0.05	0.05

REFFERENCE

REF : C

Estimated duct attenuation (in dB per 0.3m) For a straight duct with 25 mm. inside lining of sound absorption material of approximately 24 kg/m^3 density. "D" (in mm) is the inside the diameter of a round ductor smaller inside duct dimension of an oval or rectangular duct.

D, mm.	Octave Frequency Band, Hz 63	125	250	500	1000	2000	4000	8000
FOR RECTANGULAR (OR SQUARE) DUCTS :								
100 - 225	0.4	0.6	1.5	3.0	6.0	6.0	4.0	1.8
250 - 375	0.4	0.4	1.0	2.0	4.0	3.5	2.0	1.2
400 - 525	0.4	0.3	0.8	1.4	3.0	2.5	1.6	0.8
550 - 625	0.4	0.3	0.6	1.2	2.0	1.6	1.2	0.6
700 - 825	0.4	0.3	0.4	0.9	1.6	1.4	1.0	0.6
850 - 975	0.4	0.3	0.3	0.8	1.4	1.2	0.9	0.5
1000 - 1125	0.4	0.3	0.3	0.7	1.2	1.0	0.8	0.4
1150 - 1350	0.3	0.25	0.2	0.6	1.0	0.8	0.7	0.4
1375 - 1650	0.3	0.25	0.2	0.5	0.9	0.7	0.6	0.3
1675 - 1950	0.3	0.25	0.2	0.4	0.8	0.6	0.5	0.3
1975 - 2250	0.3	0.25	0.2	0.4	0.7	0.5	0.4	0.2
> OVER 2250	0.3	0.25	0.2	0.3	0.6	0.4	0.3	0.2
FOR OVAL OR ROUND DUCTS :								
100 - 225	0.4	1.2	3.0	5.4	6.0	6.0	4.0	1.3
250 - 375	0.4	0.7	1.8	3.6	4.0	3.5	2.0	1.2
400 - 525	0.4	0.5	1.2	2.5	3.0	2.5	1.6	0.8
550 - 625	0.4	0.4	1.0	2.0	2.0	1.6	1.2	0.7
700 - 825	0.4	0.3	0.8	1.6	1.6	1.4	1.0	0.6
850 - 975	0.4	0.3	0.7	1.4	1.4	1.2	0.9	0.5
1000 - 1125	0.4	0.25	0.6	1.2	1.2	1.0	0.8	0.4
1150 - 1350	0.3	0.25	0.5	1.0	1.0	0.8	0.7	0.4
1375 - 1650	0.3	0.25	0.4	0.9	0.9	0.7	0.6	0.3
1675 - 1950	0.3	0.25	0.4	0.8	0.8	0.6	0.5	0.3
1975 - 2250	0.3	0.25	0.3	0.7	0.7	0.5	0.4	0.2
>OVER 2250	0.3	0.25	0.3	0.6	0.6	0.4	0.3	0.2

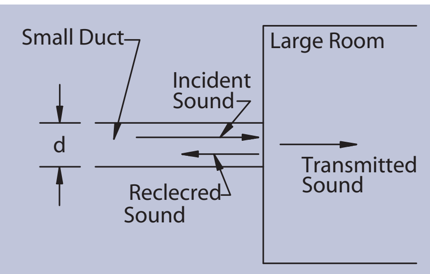
REF : D

Estimated insertion (in dB) provied by 90 bends in ducts. "D" is inside the diameter of a round duct or smaller inside duct dimension of an oval or rectangular duct.

D, mm.	Octave Frequency Band, Hz 63	125	250	500	1000	2000	4000	8000
FOR RADIUSED BEND OR BEND WITH TURNING VANES (NO LINING) :								
125 - 325	0	0	0	0	1	2	3	3
350 - 650	0	0	0	1	2	3	3	3
675 - 1075	0	0	1	2	3	3	3	3
1100 - 2250	0	0	1	2	3	3	3	3
> Over 2250	0	0	1	2	3	3	3	3
FOR LINED RADIUSED BEND (WITH OR WITHOUT TURNING VANES) OR FOR LINED SQUARE BEND WITH TURINING VANES, FOLLWED BY "3D" LENGTH OF DUCT LINING								
125 - 325	0	0	1	1	2	3	4	4
350 - 650	0	0	1	2	3	4	4	5
675 - 1075	0	1	2	3	4	4	5	5
1100 - 2250	1	2	3	4	4	5	5	6
>Over 2250	0	3	4	4	5	5	6	6
FOR LINED SQUARE BEND WITHOUT TURNING VANES, FOLLOWED BY "3D" LENGTH OF DUCT LINING :								
125 - 325	0	0	1	2	3	4	6	8
350 - 650	0	1	2	3	4	6	8	10
675 - 1075	1	2	3	4	5	6	8	10
1100 - 2250	2	3	4	5	6	8	10	12
>Over 2250	3	4	5	6	8	10	12	12

REF : E

End reflection effect (power loss in dB) of an open duct termination in a room.

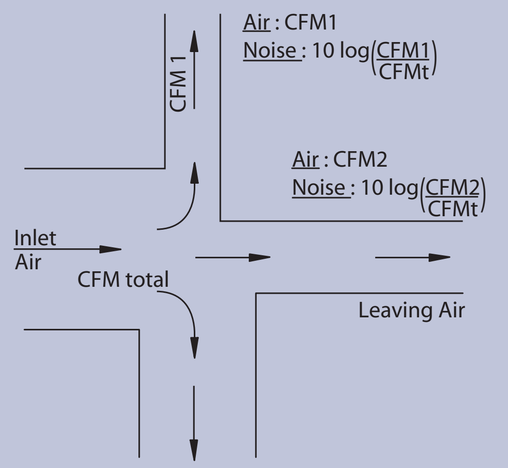


Duct Dimension d,mm.	End Reflection Effect (Loss, in dB*) Octave Frequency Band, Hz					
	31	63	125	250	500	1000
125	17	12	8	4	1	0
250	12	8	4	1	0	0
500	8	4	1	0	0	0
1000	4	1	0	0	0	0
2000	1	0	0	0	0	0

* These losses apply generally for all types and locations of duct dimensions into a room.

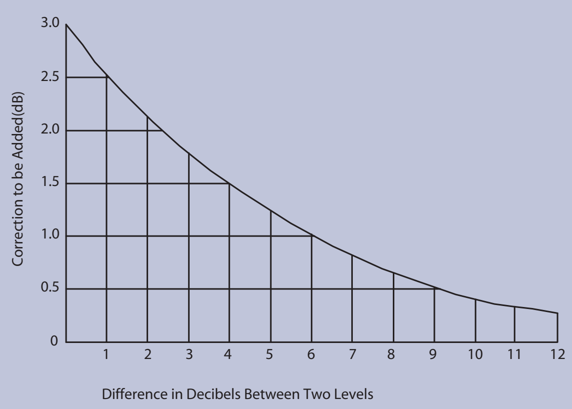
REF : F

Sample calculation showing sound power distribution in the ducts leaving a duct division.



REF : G

Chart for adding dB.



REFFERENCE

Multiply	By	To Obtain
inches of water (i.w.g.) or (in H_2O) @ 39.2 °F	2.491	millibars (mb)
	1.868	millimeters mercury (mmHg)
	25.4	kilograms per square meter (N/m ²)
	249.1	Newtons per square meter (N/m²)
	249.1	Pascal (Pa)
	0.03613	Pounds force per square in (psl)
kilograms (kg)	2.205	pounds mass (lb)
kilograms per cubic meter (kg/m³)	0.06243	pounds mass per cubic ft (lb/ft³)
kilograms per cubic meter (kg/m³)	0.2048	pounds mass per square ft (lb/ft²)
meters (m)	100	centimeters (cm)
	3.281	feet (ft)
	39.37	inches (in.)
	10^3	millimeters (mm)
	1.0936	yards (yd)
meters per second (m/s)	196.85	feet per minute (fpm)
pounds mass (lb)	0.4535	kilograms (kg)
pounds mass per cubic ft (lb/ft³)	16.02	kilograms per cubic meter (kg/m³)
pounds mass per square ft (lb/ft²)	4.882	kilogramsper square meter (kg/m²)
pounds mass per square inch (lb/inch²)	0.0703	kilograms per square centimeter (kg/cm²)
square feet (ft²)	929	square centimeters (cm²)
	144	square inches (in²)
	0.0929	square meters (m²)
square inches (in.²)	6.452	square centimeters (cm ²)
	6.994×10^{-3}	square feet (ft²)
	645.2	square millimeters (mm²)
square meters (m²)	10.76	square feet (ft ²)
	1550	square inches (in²)
atmospheres	760	mm Mercury at 1 ºc
	29.92	inches of Mercury at 4 ºc
	14.696	pounds force per square inch (psi or lb/in. ²)
centimeters (cm)	0.03281	feet (ft)
	0.3937	inches (in.)
	0.01	meters (m)
	10	millimeters (mm)
	0.010936	yards (yd)
centimeter per second (cm/s)	1.969	feet per minte (ft/min)
cubic feet (ft³)	0.02832	cubic meters (m³)
	28.32	liter (l)
cubic feet per minute (ft³/min or cim)	1.699	cubic meters per hour (m³/h)
	0.4719	liters per second (l/s.)
	4.719×10^{-4}	cubic meters per second (m³/s)
	7.4805	U.S. gallons per minute (gpm)

Multiply	By	To Obtain
cubic meters (m³)	35.31	cubic feet (ft ³)
	10^3	liters
cubic meters per hour (m³/h)	0.588	cubic feet per minute (cfm)
	0.278	liters per second (l/s)
cubic meters per hour (m³/s)	2118.88	cubic feet per minute (cfm or ft³/m)
feet (ft)	30.48	centimeters (cm)
	12	inches (in.)
	0.3048	meters (m)
	304.8	millimeters (mm)
feet per minute (fpm)	0.5080	centimeters per second (cm/s)
	0.00508	meters per scecond (m/s)
inches (in.)	2.54	centimetes (cm)
	0.0833	feet (ft)
	0.0254	meters (m)
	25.4	millimeters (mm)

U.S. Standard Cold Rolled Steel
Gages, Thickness and Weights

Gage	Thickness		Estimated Weight	
No.	in.	mm.	l b/ft ²	Kg/m²
10	0.1345	3.4163	5.625	27.46
11	0.1196	3.0378	5.00	24.41
12	0.1046	2.6568	4.375	21.36
14	0.0747	1.8974	3.125	15.26
16	0.0598	1.5189	2.5	12.21
18	0.0478	1.2141	2.0	9.764
20	0.0359	0.9119	1.50	7.323
22	0.0299	0.7595	1.26	6.10
24	0.0239	0.6671	1.00	4.882
26	0.0179	0.4547	0.75	3.66

For galvanized steel G-90 coating, added thickness = 0.0037 in. (0.094 mm), added weight = 0.156 lb/ft² (0.76 kg/m²)

U.S. Standard Cold Rolled Steel
Gages, Thickness and Weights

Gage	Thickness		Estimated Weight	
No.	in.	mm.	l b/ft ²	Kg/m²
10	0.128	3.251	5.353	26.13
11	0.116	2.946	4.849	23.67
12	0.104	2.642	4.350	21.24
14	0.080	2.032	3.347	16.34
16	0.064	1.626	2.676	13.06
18	0.048	1.219	2.009	9.80
20	0.036	0.914	1.504	7.34
22	0.028	0.711	1.171	5.71
24	0.022	0.559	0.921	4.50
26	0.018	0.457	0.754	3.68

HOW TO ORDER : TYPE LF

EXAMPLE 1,200 X 1,000 MF X 1,500
Duct Width x Duct Height x Model x Duct Length

Abie Tiger Acoustical Silencers

Silencer Splitters

Splitter is normally manufactured from minimum gauge 24 perforated galvanized steel with 2.6 mm hole diameter giving a minimum 17.23% open area. The bul nose in the machine side is fabricated from minimum gauge 24 galvanized steell, the bull nose at the other end is manufactured from same material as splitter. The splitters are inserted and locked into the duct silencer casing by an interlocking track. The track is from minimum gauge 24 solid galvanized steel. The infil material is absorption material tested in accordance with ASTM C423, ISO 354/1995 E in the reverberation chamber. The material is compressed to not less than 10% of its volume to ensure solid packing in the splitter.

Outer Casing

The outer casing is manufactured from minimum gauge 20 galvanized steel.

Silencer Performance

The acoustic silencer system is normally supplied as the module. Each module has been tested in according to ASTM E477-90a, BS 4718 in the reverberation laboratory for dynamic insertion loss, regenerated noise and air flow data. The silencer has been tested at the Warrington Fire Research Laboratories in Melbourne Australia and found to exceed the test requirements of BS 476 Part 24. This standard requires the silencer to be tested at a temperature of 650°C of 60 minutes and suffer no damage, distortion or buckling.

SELF-NOISE SOUND POWER LEVEL, dB 10^{-12} WATES

Mode	Silencer	Octave Band Center Frequency, Hz							
	Face Velocity m/s	63	125	250	500	1000	2000	4000	8000
	10 Supply	58	49	49	48	46	48	46	37
	8 Supply	50	43	46	40	38	45	36	29
	6 Supply	40	30	39	31	33	37	22	22
LF	3 Supply	28	20	23	20	20	23	<20	<20
	3 Return	30	25	28	22	39	41	31	<20
	6 Return	42	43	42	40	50	50	40	22
	8 Return	50	48	48	47	52	53	46	35
	10 Return	53	54	53	55	55	55	53	42

Supply : Air traveling in the same direction of sound propagation.
Return : Air traveling in the opposite direction of the sound propagation (Interpolate self-noise for intermediate velocities.)

DYNAMIC INSERTION LOSS (DIL) RATINGS. (dB)

Model	Length (mm.)	Silencer Face Velocity m/s	Octave Band Center Frequency, (Hz) 63	125	250	500	1000	2000	4000	8000
		10 Supply	3	3	5	8	14	17	10	7
		8 Supply	3	3	5	8	14	17	10	7
		6 Supply	3	3	6	9	15	17	9	7
		3 Supply	3	3	6	9	15	17	9	7
600LF	600	0 Static	3	4	7	9	15	13	9	7
		3 Return	3	5	7	10	16	13	8	7
		6 Return	3	5	7	10	16	13	8	7
		8 Return	3	5	8	11	17	12	7	7
		10 Return	3	5	8	11	17	12	7	7
		10 Supply	3	4	8	13	22	25	15	11
		8 Supply	3	4	8	13	22	25	15	11
		6 Supply	3	4	9	13	22	25	15	11
		3 Supply	3	4	9	13	22	25	15	11
900LF	900	0 Static	4	6	10	14	23	22	14	11
		3 Return	4	6	11	15	24	22	14	11
		6 Return	4	6	11	15	24	20	14	11
		8 Return	5	6	11	15	25	20	14	11
		10 Return	5	6	11	16	25	18	14	11
		10 Supply	5	5	10	15	26	32	17	13
		8 Supply	5	5	10	15	26	32	17	13
		6 Supply	5	6	11	16	26	31	16	13
		3 Supply	5	6	11	16	25	31	16	13
1200LF	1200	0 Static	6	7	12	17	27	28	15	12
		3 Return	7	8	13	18	27	25	15	12
		6 Return	7	8	13	18	27	25	15	12
		8 Return	7	8	14	19	28	24	15	12
		10 Return	7	8	14	19	28	24	15	12
		10 Supply	6	6	12	22	30	38	19	15
		8 Supply	6	6	12	22	30	38	19	15
		6 Supply	6	7	13	22	30	37	18	14
		3 Supply	6	7	13	23	30	37	18	14
1500LF	1500	0 Static	7	8	14	23	31	35	17	13
		3 Return	8	9	14	23	31	34	17	13
		6 Return	8	9	14	23	31	34	17	13
		8 Return	8	9	15	24	32	34	16	13
		10 Return	8	9	15	24	32	34	16	13

Model	Length (mm.)	Silencer Face Velocity m/s	63	125	250	500	1000	2000	4000	8000
		10 Supply	7	9	15	22	35	43	24	16
		8 Supply	7	9	15	22	35	43	24	16
		6 Supply	7	10	16	23	35	42	23	16
		3 Supply	7	10	16	23	35	42	23	16
1800LF	1800	0 Static	8	11	17	24	36	40	22	15
		3 Return	9	12	18	25	36	37	21	15
		6 Return	9	12	18	25	36	37	21	15
		8 Return	9	13	18	26	37	35	20	14
		10 Return	9	13	18	26	37	35	20	14
		10 Supply	8	12	17	27	39	47	28	17
		8 Supply	8	12	17	27	39	47	28	17
		6 Supply	8	13	17	27	40	46	27	17
		3 Supply	8	13	18	27	40	46	27	17
2100LF	2100	0 Static	8	14	18	28	41	42	26	16
		3 Return	10	15	19	29	42	42	25	16
		6 Return	10	15	19	29	42	42	25	16
		8 Return	10	15	19	29	42	40	25	16
		10 Return	10	15	20	29	42	40	25	16
		10 Supply	9	13	20	31	42	48	31	19
		8 Supply	9	13	20	31	42	48	31	19
		6 Supply	9	14	20	31	42	47	30	19
		3 Supply	9	14	20	31	42	47	30	19
2400LF	2400	0 Static	9	15	21	32	43	45	29	19
		3 Return	11	16	22	33	44	44	28	19
		6 Return	11	16	22	33	44	44	28	19
		8 Return	11	17	22	33	44	43	27	19
		10 Return	11	17	22	33	44	43	27	19
		10 Supply	10	17	23	39	47	49	36	23
		8 Supply	10	17	23	39	47	49	36	23
		6 Supply	10	17	23	39	47	48	36	23
		3 Supply	10	17	24	40	48	48	36	23
3000LF	3000	0 Static	10	18	24	40	48	48	35	23
		3 Return	12	18	24	40	48	48	35	23
		6 Return	12	19	25	41	49	48	35	23
		8 Return	12	20	25	41	49	48	34	23
		10 Return	12	20	25	41	50	48	33	22

Octave Band Center Frequency, (Hz)

Note :
Supply : Air is travelling in the same direction of sound propagation.
Return : Air is travelling in the opposite direction of sound propagation.
Interpolate dynamic insertion loss (DIL) from intermediate velocities.

LOW-PRESSURE DROP & FREQUENCY NOISE

TYPE LF

Duct Silencer Face Area (SQ.m.)	0.045	0.09	0.18	0.36	0.72	1.44	2.89	5.77	11.54
Self Noise Correction Factor, dB	-9	-6	-3	0	+3	+6	+9	+12	+15

Silencer Length (mm)	Static Pressure Drop, (Pa)											
600	0.4	0.9	1.6	2.6	3.7	6.6	10.3	14.8	20.1	26.3	33.3	41.1
900	0.5	1.1	1.9	2.9	4.2	7.5	12	16.9	23	30.1	38.1	47
1200	0.5	1.2	2.1	3.3	4.8	8.5	13.5	19.1	25.9	33.9	42.9	52.9
1500	0.6	1.3	2.4	3.7	5.3	9.4	14.7	21.2	28.8	37.6	47.6	58.8
1800	0.6	1.5	2.6	4.1	5.8	10.4	16.2	23.4	31.8	41.5	52.6	64.9
2100	0.7	1.6	2.8	4.4	6.4	11.3	17.7	25.5	34.7	45.3	57.3	70.8
2400	0.8	1.7	3.1	4.8	6.9	12.2	19.1	27.5	37.5	49.0	62.0	76.5
3000	0.8	1.9	3.3	5.2	7.4	13.2	20.6	29.7	40.4	52.8	66.8	82.5

Module Size width & Height (mm) x (mm)	Weight in kg per liner meter	Face Area sq.m.	Air Flow (cu.m./s.)											
300 x 300	20	0.09	0.09	0.14	0.18	0.23	0.27	0.36	0.45	0.54	0.63	0.72	0.81	0.90
300 x 450	30	0.14	0.14	0.20	0.27	0.34	0.41	0.54	0.68	0.81	0.95	1.08	1.22	1.35
300 x 600	40	0.18	0.18	0.27	0.36	0.45	0.54	0.72	0.90	1.08	1.26	1.44	1.62	1.80
300 x 750	50	0.23	0.23	0.34	0.45	0.56	0.68	0.90	1.13	1.35	1.58	1.80	2.03	2.25
300 x 900	60	0.27	0.27	0.41	0.54	0.68	0.81	1.08	1.35	1.62	1.89	2.16	2.43	2.70
300 x 1,050	70	0.32	0.32	0.47	0.63	0.79	0.95	1.26	1.58	1.89	2.21	2.52	2.84	3.15
300 x 1,200	80	0.36	0.36	0.54	0.72	0.90	1.08	1.44	1.80	2.16	2.52	2.88	3.24	3.60
600 x 450	60	0.27	0.27	0.41	0.54	0.68	0.81	1.08	1.35	1.62	1.89	2.16	2.43	2.70
600 x 600	80	0.36	0.36	0.54	0.72	0.90	1.08	1.44	1.80	2.16	2.52	2.88	3.24	3.60
600 x 750	100	0.45	0.45	0.68	0.90	1.13	1.35	1.80	2.25	2.70	3.15	3.60	4.05	4.50
600 x 900	120	0.54	0.54	0.81	1.08	1.35	1.62	2.16	2.70	3.24	3.78	4.32	4.86	5.40
600 x 1,050	140	0.63	0.63	0.95	1.26	1.58	1.89	2.52	3.15	3.78	4.41	5.04	5.67	6.30
600 x 1,200	160	0.72	0.72	1.08	1.44	1.80	2.16	2.88	3.60	4.32	5.04	5.76	6.48	7.20
1,200 x 750	200	0.90	0.90	1.35	1.80	2.25	2.70	3.60	4.50	5.40	6.30	7.20	8.10	9.00
1,200 x 900	240	1.08	1.08	1.62	2.16	2.70	3.24	4.32	5.40	6.48	7.56	8.64	9.72	10.80
1,200 x 1,050	280	1.26	1.26	1.89	2.52	3.15	3.78	5.04	6.30	7.56	8.82	10.08	11.34	12.60
1,200 x 1,200	320	1.44	1.44	2.16	2.88	3.60	4.32	5.76	7.20	8.64	10.08	11.52	12.96	14.40
Silencer entering Face Velocity, (m/s)			1	1.5	2	2.5	3	4	5	6	7	8	9	10

Note :

-If sound attenuators are installed immediately right after or before transitions, elbows, or at the intake or discharge of the system, the pressure drops must be changed from those given above by at least a factor at 1.5 times, depending upon specific conditions.

- Dynamic insertion loss and air flow data has been certified in accordance with BS 4718 and ASTM E477.

- All acoustic data has been determined in the Abie Tiger Acoustics Aero-Dynamic Reverberation Laboratory in a reverberant room of volume 332.6 m^3.

- All data was obtained on duct sizes 610 mm. x 610 mm.

HOW TO ORDER : TYPE LMF

EXAMPLE 1,200 X 1,000 LMF X 1,500
Duct Width x Duct Height x Model x Duct Length

Abie Tiger Acoustical Silencers

Silencer Splitters

Splitter is normally manufactured from minimum gauge 24 perforated galvanized steel with 2.6 mm hole diameter giving a minimum 17.23% open area. The bul nose in the machine side is fabricated from minimum gauge 24 galvanized steell, the bull nose at the other end is manufactured from same material as splitter. The splitters are inserted and locked into the duct silencer casing by an interlocking track. The track is from minimum gauge 24 solid galvanized steel. The infil material is absorption material tested in accordance with ASTM C423, ISO 354/1995 E in the reverberation chamber. The material is compressed to not less than 10% of its volume to ensure solid packing in the splitter.

Outer Casing

The outer casing is manufactured from minimum gauge 20 galvanized steel.

Silencer Performance

The acoustic silencer system is normally supplied as the module. Each module has been tested in according to ASTM E477-90a, BS 4718 in the reverberation laboratory for dynamic insertion loss, regenerated noise and air flow data. The silencer has been tested at the Warrington Fire Research Laboratories in Melbourne Australia and found to exceed the test requirements of BS 476 Part 24. This standard requires the silencer to be tested at a temperature of 650°C of 60 minutes and suffer no damage, distortion or buckling.

SELF-NOISE SOUND POWER LEVEL, dB 10^{-12} WATES

Model	Silencer	Octave Band Center Frequency, Hz							
	Face Velocity m/s	**63**	**125**	**250**	**500**	**1000**	**2000**	**4000**	**8000**
	10 Supply	52	50	60	58	53	54	55	48
	8 Supply	51	43	47	48	47	45	42	35
	6 Supply	49	36	34	37	40	39	29	22
LMF	3 Supply	30	22	23	28	31	29	<20	<20
	3 Return	29	25	37	35	35	34	28	<20
	6 Return	47	39	46	48	49	44	39	29
	8 Return	50	45	50	50	52	53	50	41
	10 Return	54	51	52	53	54	61	60	52

Supply : Air traveling in the same direction of sound propagation.
Return : Air traveling in the opposite direction of the sound propagation (Interpolate self-noise for intermediate velocities.)

DYNAMIC INSERTION LOSS (DIL) RATINGS. (dB)

Model	Length (mm.)	Silencer Face Velocity m/s	63	125	250	500	1000	2000	4000	8000
			Octave Band Center Frequency, (Hz)							
		10 Supply	3	5	10	13	12	8	7	7
		8 Supply	3	5	10	13	12	8	7	7
		6 Supply	3	5	10	13	12	8	7	7
		3 Supply	3	5	10	13	12	8	7	7
600LMF	600	0 Static	3	5	10	13	12	9	7	7
		3 Return	4	5	11	13	12	9	7	7
		6 Return	4	5	11	13	12	9	7	7
		8 Return	5	6	11	14	12	9	7	7
		10 Return	5	6	11	14	12	9	7	7
		10 Supply	4	6	14	19	18	11	10	9
		8 Supply	4	6	14	19	18	11	10	9
		6 Supply	4	6	14	20	18	11	10	9
		3 Supply	4	7	15	20	18	12	10	10
900LMF	900	0 Static	4	7	15	20	18	12	10	10
		3 Return	5	7	16	21	19	12	10	10
		6 Return	5	7	16	21	19	12	11	11
		8 Return	6	8	17	22	19	13	11	11
		10 Return	6	9	17	22	20	13	12	11
		10 Supply	7	11	20	23	23	14	11	10
		8 Supply	7	11	20	24	23	15	11	10
		6 Supply	8	11	20	24	23	14	11	10
		3 Supply	8	11	20	25	23	14	11	11
1200LMF	1200	0 Static	8	12	21	25	23	14	11	11
		3 Return	10	12	21	26	24	15	12	11
		6 Return	10	12	21	26	24	15	12	11
		8 Return	10	12	22	27	24	15	12	12
		10 Return	10	13	22	27	24	15	13	12
		10 Supply	8	14	24	27	25	16	12	11
		8 Supply	8	14	24	28	26	17	12	11
		6 Supply	8	14	25	28	26	16	12	11
		3 Supply	9	14	25	30	27	16	12	12
1500LMF	1500	0 Static	9	15	26	31	27	15	12	12
		3 Return	11	15	26	31	28	17	13	12
		6 Return	11	16	26	32	28	16	13	12
		8 Return	11	16	28	32	29	17	13	13
		10 Return	11	17	28	33	31	18	14	14

RECTANGULAR DUCT SILENCER
LOW-MID FREQUENCY NOISE ATTENUATION

TYPE
LMF

Abie Tiger

Model	Length (mm.)	Silencer Face Velocity m/s	63	125	250	500	1000	2000	4000	8000
			Octave Band Center Frequency, (Hz)							
		10 Supply	12	15	27	36	33	19	14	13
		8 Supply	12	15	27	36	33	19	14	14
		6 Supply	12	15	28	36	33	19	14	14
		3 Supply	12	15	28	37	33	19	14	14
1800LMF	1800	0 Static	12	16	29	37	34	18	14	14
		3 Return	13	17	29	36	35	19	15	14
		6 Return	13	17	29	36	35	19	15	14
		8 Return	13	17	30	37	35	19	15	14
		10 Return	13	18	30	36	35	19	15	15
		10 Supply	14	16	28	38	39	20	15	15
		8 Supply	14	16	28	38	39	20	15	15
		6 Supply	14	17	29	39	39	20	15	15
		3 Supply	14	17	29	39	40	20	15	15
2100LMF	2100	0 Static	16	18	30	40	40	20	15	15
		3 Return	16	18	30	40	40	20	16	16
		6 Return	16	19	31	41	40	22	16	16
		8 Return	16	19	31	42	40	22	17	16
		10 Return	16	20	32	43	40	23	18	16
		10 Supply	15	17	32	44	42	23	18	16
		8 Supply	15	17	32	45	42	23	18	16
		6 Supply	15	18	33	44	43	23	18	16
		3 Supply	15	19	34	44	42	23	17	16
2400LMF	2400	0 Static	15	20	35	45	43	23	17	16
		3 Return	17	20	35	45	44	23	17	17
		6 Return	17	20	36	46	44	23	18	17
		8 Return	17	21	36	46	44	23	18	17
		10 Return	17	21	36	46	44	25	19	17
		10 Supply	17	20	37	49	49	30	21	17
		8 Supply	17	20	37	49	49	30	21	17
		6 Supply	17	21	38	49	49	29	20	17
		3 Supply	17	21	40	49	50	29	20	17
3000LMF	3000	0 Static	17	23	41	50	50	28	20	17
		3 Return	18	23	42	51	50	28	20	18
		6 Return	18	24	42	51	52	27	20	18
		8 Return	19	25	42	52	52	27	19	18
		10 Return	21	26	43	53	52	27	19	18

Note :
Supply : Air is travelling in the same direction of sound propagation.
Return : Air is travelling in the opposite direction of sound propagation.
Interpolate dynamic insertion loss (DIL) from intermediate velocities.

Abie Tiger

RECTANGULAR DUCT SILENCER LOW-MID FREQUENCY NOISE ATTENUATION

TYPE LMF

Duct Silencer Face Area (SQ.m.)	0.045	0.09	0.18	0.36	0.72	1.44	2.89	5.77	11.54
Self Noise Correction Factor, dB	-9	-6	-3	0	+3	+6	+9	+12	+15

Silencer Length (mm)	Static Pressure Drop, (Pa)											
600	1.8	4.1	7.2	11.3	16.2	28.8	45.0	64.8	88.2	115.2	145.8	180.0
900	1.9	4.3	7.7	12	17.3	30.7	49	69.1	94.1	122.9	155.6	188
1200	2.0	4.4	7.8	12.3	17.6	31.4	50.0	70.6	96.0	125.4	158.8	196.0
1500	2.0	4.6	8.1	12.7	18.3	32.5	50.8	73.1	99.5	129.9	164.4	203.0
1800	2.1	4.7	8.4	13.1	18.9	33.6	52.5	75.6	102.9	134.3	170.0	209.9
2100	2.2	4.9	8.7	13.6	19.5	34.7	54.3	78.1	106.3	138.9	175.8	217.0
2400	2.2	5.0	9.0	14.0	20.2	35.8	56.0	80.6	109.8	143.4	181.4	224.0
3000	2.3	5.2	9.3	14.5	20.9	37.1	58.0	83.5	113.7	148.5	187.9	232.0

Module Size width & Height (mm) x (mm)	Weight in kg per liner meter	Face Area sq.m.	Air Flow (cu.m./s.)											
300 x 300	20	0.09	0.09	0.14	0.18	0.23	0.27	0.36	0.45	0.54	0.63	0.72	0.81	0.90
300 x 450	30	0.14	0.14	0.20	0.27	0.34	0.41	0.54	0.68	0.81	0.95	1.08	1.22	1.35
300 x 600	40	0.18	0.18	0.27	0.36	0.45	0.54	0.72	0.90	1.08	1.26	1.44	1.62	1.80
300 x 750	50	0.23	0.23	0.34	0.45	0.56	0.68	1.90	1.13	1.35	1.58	1.80	2.03	2.25
300 x 900	60	0.27	0.27	0.41	0.54	0.68	0.81	1.08	1.35	1.62	1.89	2.16	2.43	2.70
300 x 1,050	70	0.32	0.32	0.47	0.63	0.79	0.95	1.26	1.58	1.89	2.21	2.52	2.84	3.15
300 x 1,200	80	0.36	0.36	0.54	0.72	0.90	1.08	1.44	1.80	2.16	2.52	2.88	3.24	3.60
600 x 450	60	0.27	0.27	0.41	0.54	0.68	0.81	1.08	1.35	1.62	1.89	2.16	2.43	2.70
600 x 600	80	0.36	0.36	0.54	0.72	0.90	1.08	1.44	1.80	2.16	2.52	2.88	3.24	3.60
600 x 750	100	0.45	0.45	0.68	0.90	1.13	1.35	1.80	2.25	2.70	3.15	3.60	4.05	4.50
600 x 900	120	0.54	0.54	0.81	1.08	1.35	1.62	2.16	2.70	3.24	3.78	4.32	4.86	5.40
600 x 1,050	140	0.63	0.63	0.95	1.26	1.58	1.89	2.52	3.15	3.78	4.41	5.04	5.67	6.30
600 x 1,200	160	0.72	0.72	1.08	1.44	1.80	2.16	2.88	3.60	4.32	5.04	5.76	6.48	7.20
1,200 x 750	200	0.90	0.90	1.35	1.80	2.25	2.70	3.60	4.50	5.40	6.30	7.20	8.10	9.00
1,200 x 900	240	1.08	1.08	1.62	2.16	2.70	3.24	4.32	5.40	6.48	7.56	8.64	9.72	10.80
1,200 x 1,050	280	1.26	2.26	1.89	2.52	3.15	3.78	5.04	6.30	7.56	8.82	10.08	11.34	12.60
1,200 x 1,200	320	1.44	1.44	2.16	2.88	3.60	4.32	5.76	7.20	8.74	10.08	11.52	12.96	14.40
Silencer entering Face Velocity, (m/s)			1	1.5	2	2.5	3	4	5	6	7	8	9	10

Note :

-If sound attenuators are installed immediately right after or before transitions, elbows, or at the intake or discharge of the system, the pressure drops must be changed from those given above by at least a factor at 1.5 times, depending upon specific conditions.

- Dynamic insertion loss and air flow data has been certified in accordance with BS 4718 and ASTM E477.

- All acoustic data has been determined in the Abie Tiger Acoustics Aero-Dynamic Reverberation Laboratory in a reverberant room of volume 332.6 m^3.

- All data was obtained on duct sizes 610 mm. x 610 mm.

HOW TO ORDER : TYPE MF

EXAMPLE 1,200 X 1,000 MF X 1,500
Duct Width x Duct Height x Model x Duct Length

Abie Tiger Acoustical Silencers

Silencer Splitters

Splitter is normally manufactured from minimum gauge 24 perforated galvanized steel with 2.6 mm hole diameter giving a minimum 17.23% open area. The bul nose in the machine side is fabricated from minimum gauge 24 galvanized steell, the bull nose at the other end is manufactured from same material as splitter. The splitters are inserted and locked into the duct silencer casing by an interlocking track. The track is from minimum gauge 24 solid galvanized steel. The infil material is absorption material tested in accordance with ASTM C423, ISO 354/1995 E in the reverberation chamber. The material is compressed to not less than 10% of its volume to ensure solid packing in the splitter.

Outer Casing

The outer casing is manufactured from minimum gauge 20 galvanized steel.

Silencer Performance

The acoustic silencer system is normally supplied as the module. Each module has been tested in according to ASTM E477-90a, BS 4718 in the reverberation laboratory for dynamic insertion loss, regenerated noise and air flow data. The silencer has been tested at the Warrington Fire Research Laboratories in Melbourne Australia and found to exceed the test requirements of BS 476 Part 24. This standard requires the silencer to be tested at a temperature of 650°C of 60 minutes and suffer no damage, distortion or buckling.

SELF-NOISE SOUND POWER LEVEL, dB 10^{-12} WATES

Model	Silencer	Octave Band Center Frequency, Hz							
	Face Velocity m/s	63	125	250	500	1000	2000	4000	8000
	10 Supply	65	52	50	51	45	50	49	42
	8 Supply	58	49	47	45	42	43	42	36
	6 Supply	46	33	38	32	30	30	27	22
MF	3 Supply	28	20	23	22	21	21	22	<20
	3 Return	29	32	38	33	42	32	27	<20
	6 Return	47	43	48	43	52	49	45	26
	8 Return	55	53	52	53	55	57	54	44
	10 Return	61	58	56	57	59	61	57	46

Supply : Air traveling in the same direction of sound propagation.
Return : Air traveling in the opposite direction of the sound propagation (Interpolate self-noise for intermediate velocities.)

DYNAMIC INSERTION LOSS (DIL) RATINGS. (dB)

Model	Length (mm.)	Silencer Face Velocity m/s	Octave Band Center Frequency, (Hz) 63	125	250	500	1000	2000	4000	8000
		10 Supply	3	3	9	9	15	15	12	7
		8 Supply	3	3	9	9	15	15	12	7
		6 Supply	3	4	9	10	15	15	11	6
		3 Supply	3	4	9	11	15	15	11	6
600MF	600	0 Static	3	5	9	11	16	14	11	6
		3 Return	4	5	9	13	17	13	10	5
		6 Return	4	5	9	13	17	13	9	5
		8 Return	4	5	9	13	17	13	9	5
		10 Return	4	5	9	13	17	13	8	5
		10 Supply	3	6	13	18	22	22	19	10
		8 Supply	3	6	13	18	22	22	19	10
		6 Supply	4	6	13	18	23	22	17	9
		3 Supply	4	6	13	18	23	22	17	9
900MF	900	0 Static	5	7	13	20	24	21	16	9
		3 Return	6	7	13	20	25	20	15	8
		6 Return	6	7	13	22	25	20	14	8
		8 Return	6	7	13	22	26	19	13	7
		10 Return	7	7	13	22	26	19	13	7
		10 Supply	5	7	15	21	32	28	21	12
		8 Supply	5	7	15	21	32	28	21	12
		6 Supply	6	8	15	22	32	28	20	11
		3 Supply	6	8	15	23	32	28	20	11
1200MF	1200	0 Static	7	9	15	24	33	27	19	11
		3 Return	7	9	15	25	33	26	18	10
		6 Return	7	10	15	25	33	26	18	10
		8 Return	8	10	15	26	34	25	17	9
		10 Return	8	10	15	26	34	25	17	9
		10 Supply	6	9	17	31	41	35	24	13
		8 Supply	6	9	17	31	41	35	24	13
		6 Supply	7	9	17	31	41	35	24	13
		3 Supply	7	9	17	31	41	34	22	13
1500MF	1500	0 Static	8	10	17	32	41	33	22	12
		3 Return	8	10	17	32	41	33	22	11
		6 Return	8	11	17	32	41	33	22	11
		8 Return	9	11	17	33	41	32	21	11
		10 Return	9	12	17	33	41	32	21	10

RECTANGULAR DUCT SILENCER
MID-FREQUENCY NOISE ATTENUATION

Abie Tiger

TYPE MF

Model	Length (mm.)	Silencer Face Velocity m/s	63	125	250	500	1000	2000	4000	8000
			Octave Band Center Frequency, (Hz)							
		10 Supply	8	10	23	32	43	39	27	16
		8 Supply	8	10	23	32	43	39	27	16
		6 Supply	9	11	23	33	44	38	26	16
		3 Supply	9	11	22	33	44	38	26	16
1800MF	1800	0 Static	9	12	22	34	44	37	25	14
		3 Return	10	12	22	34	45	36	24	12
		6 Return	10	13	21	34	45	36	24	12
		8 Return	10	13	21	35	45	35	24	12
		10 Return	10	13	20	35	45	35	23	12
		10 Supply	9	13	23	35	46	43	30	17
		8 Supply	9	13	23	35	46	43	30	17
		6 Supply	10	13	23	35	46	43	29	17
		3 Supply	10	14	24	36	48	43	29	16
2100MF	2100	0 Static	10	14	24	36	48	42	28	16
		3 Return	11	15	24	36	49	40	28	15
		6 Return	11	15	24	37	49	40	28	14
		8 Return	11	16	25	37	50	39	27	13
		10 Return	11	16	25	37	50	39	27	13
		10 Supply	11	14	28	37	47	47	35	19
		8 Supply	11	14	28	37	47	47	35	19
		6 Supply	11	14	28	38	48	46	34	19
		3 Supply	11	14	28	38	49	46	34	19
2400MF	2400	0 Static	11	16	28	39	50	45	32	17
		3 Return	12	16	28	39	50	43	29	15
		6 Return	12	16	27	39	50	43	29	15
		8 Return	12	17	27	40	51	42	29	15
		10 Return	12	17	26	40	51	42	28	15
		10 Supply	13	19	32	40	50	51	44	23
		8 Supply	13	19	32	40	50	51	44	23
		6 Supply	13	20	32	41	51	51	43	22
		3 Supply	13	20	33	41	51	51	43	22
3000MF	3000	0 Static	13	20	33	42	52	50	40	21
		3 Return	14	21	34	43	53	49	36	21
		6 Return	14	21	34	43	53	49	36	20
		8 Return	14	22	34	45	54	48	35	20
		10 Return	14	22	34	45	54	48	35	19

Note :
Supply : Air is travelling in the same direction of sound propagation.
Return : Air is travelling in the opposite direction of sound propagation.
Interpolate dynamic insertion loss (DIL) from intermediate velocities.

RECTANGULAR DUCT SILENCER MID-FREQUENCY NOISE ATTENUATION

TYPE MF

Duct Silencer Face Area (SQ.m.)	0.045	0.09	0.18	0.36	0.72	1.44	2.89	5.77	11.54
Self Noise Correction Factor, dB	-9	-6	-3	0	+3	+6	+9	+12	+15

Silencer Length (mm)	Static Pressure Drop, (Pa)											
600	0.9	1.9	3.4	5.3	7.7	13.6	21.3	30.6	41.7	54.4	68.9	85.0
900	1	2.2	3.9	6.1	8.8	15.7	25	35.3	48	62.7	79.4	97
1200	1.1	2.5	4.4	6.9	9.9	17.6	28.0	39.5	53.8	70.2	88.9	109.8
1500	1.2	2.7	4.9	7.6	11.0	19.5	30.5	43.9	59.8	78.1	98.8	122
1800	1.3	3.0	5.4	8.4	12.1	21.4	33.5	48.2	65.7	85.8	108.5	134
2100	1.5	3.3	5.8	9.1	13.1	23.4	36.5	52.6	71.5	93.4	118.3	146
2400	1.6	3.6	6.3	9.9	14.2	25.3	39.5	56.9	77.4	101.1	128.0	158
3000	1.7	3.8	6.8	10.6	15.3	27.2	42.5	61.2	83.3	108.8	137.7	170

Module Size width & Height (mm) x (mm)	Weight in kg per liner meter	Face Area sq.m.	Air Flow (cu.m./s.)											
300 x 300	20	0.09	0.09	0.14	0.18	0.23	0.27	0.36	0.45	0.54	0.63	0.72	0.81	0.90
300 x 450	30	0.14	0.14	0.20	0.27	0.34	0.41	0.54	0.68	0.81	0.95	1.08	1.22	1.35
300 x 600	40	0.18	0.18	0.27	0.36	0.45	0.54	0.72	0.90	1.08	1.26	1.44	1.62	1.80
300 x 750	50	0.23	0.23	0.34	0.45	0.56	0.68	0.90	1.13	1.35	1.58	1.80	2.03	2.25
300 x 900	60	0.27	0.27	0.41	0.54	0.68	0.81	1.08	1.35	1.62	1.89	2.16	2.43	2.70
300 x 1,050	70	0.32	0.32	0.47	0.63	0.79	0.95	1.26	1.58	1.89	2.21	2.52	2.84	3.15
300 x 1,200	80	0.36	0.36	0.54	0.72	0.90	1.08	1.44	1.80	2.16	2.52	2.88	3.24	3.60
600 x 450	60	0.27	0.27	0.41	0.54	0.68	0.81	1.08	1.35	1.62	1.89	2.16	2.43	2.70
600 x 600	80	0.36	0.36	0.54	0.72	0.90	1.08	1.44	1.80	2.16	2.52	2.88	3.24	3.60
600 x 750	100	0.45	0.45	0.68	0.90	1.13	1.35	1.80	2.25	2.70	3.15	3.60	4.05	4.50
600 x 900	120	0.54	0.54	0.81	1.08	1.35	1.62	2.16	2.70	3.24	3.78	4.32	4.86	5.40
600 x 1,050	140	0.63	0.63	0.95	1.26	1.58	1.89	2.52	3.15	3.78	4.41	5.04	5.67	6.30
600 x 1,200	160	0.72	0.72	1.08	1.44	1.80	2.16	2.88	3.60	4.32	5.04	5.76	6.48	7.20
1,200 x 750	200	0.9	0.9	1.35	1.80	2.25	2.70	3.60	4.50	5.40	6.30	7.20	8.10	9.00
1,200 x 900	240	1.08	1.08	1.62	2.16	2.70	3.24	4.32	5.40	6.48	7.56	8.64	9.72	10.80
1,200 x 1,050	280	1.26	1.26	1.89	2.52	3.15	3.78	5.04	6.30	7.56	8.82	10.08	11.34	12.60
1,200 x 1,200	320	1.44	1.44	2.16	2.88	3.60	4.32	5.76	7.20	8.64	10.08	11.52	12.96	14.40
Silencer entering Face Velocity, (m/s)			1	1.5	2	2.5	3	4	5	6	7	8	9	10

Note :

-If sound attenuators are installed immediately right after or before transitions, elbows, or at the intake or discharge of the system, the pressure drops must be changed from those given above by at least a factor at 1.5 times, depending upon specific conditions.

- Dynamic insertion loss and air flow data has been certified in accordance with BS 4718 and ASTM E477.

- All acoustic data has been determined in the Abie Tiger Acoustics Aero-Dynamic Reverberation Laboratory in a reverberant room of volume 332.6 m^3.

- All data was obtained on duct sizes 610 mm. x 610 mm.

HOW TO ORDER : TYPE ELF

EXAMPLE 1,200 X 1,000 ELF X 1,500
Duct Width x Duct Height x Model x Duct Length

Abie Tiger Acoustical Silencers

Silencer Splitters

Splitter is normally manufactured from minimum gauge 24 perforated galvanized steel with 2.6 mm hole diameter giving a minimum 17.23% open area. The bul nose in the machine side is fabricated from minimum gauge 24 galvanized steell, the bull nose at the other end is manufactured from same material as splitter. The splitters are inserted and locked into the duct silencer casing by an interlocking track. The track is from minimum gauge 24 solid galvanized steel. The infil material is absorption material tested in accordance with ASTM C423, ISO 354/1995 E in the reverberation chamber. The material is compressed to not less than 10% of its volume to ensure solid packing in the splitter.

Outer Casing

The outer casing is manufactured from minimum gauge 20 galvanized steel.

Silencer Performance

The acoustic silencer system is normally supplied as the module. Each module has been tested in according to ASTM E477-90a, BS 4718 in the reverberation laboratory for dynamic insertion loss, regenerated noise and air flow data. The silencer has been tested at the Warrington Fire Research Laboratories in Melbourne Australia and found to exceed the test requirements of BS 476 Part 24. This standard requires the silencer to be tested at a temperature of 650°C of 60 minutes and suffer no damage, distortion or buckling.

SELF-NOISE SOUND POWER LEVEL, dB 10^{-12} WATES

Model	Silencer	Octave Band Center Frequency, Hz							
	Face Velocity m/s	**63**	**125**	**250**	**500**	**1000**	**2000**	**4000**	**8000**
	10 Supply	69	65	66	67	63	62	63	62
	8 Supply	54	56	58	54	50	57	59	53
	6 Supply	48	41	43	45	46	51	46	39
ELF	3 Supply	29	26	31	27	33	41	31	24
	3 Return	36	34	36	32	34	41	32	24
	6 Return	47	44	44	41	47	51	46	38
	8 Return	53	51	55	55	55	60	62	56
	10 Return	59	52	56	62	64	64	66	64

Supply : Air traveling in the same direction of sound propagation.
Return : Air traveling in the opposite direction of the sound propagation (Interpolate self-noise for intermediate velocities.)

DYNAMIC INSERTION LOSS (DIL) RATINGS. (dB)

Model	Length (mm.)	Silencer Face Velocity m/s	Octave Band Center Frequency, (Hz) 63	125	250	500	1000	2000	4000	8000
		10 Supply	5	8	14	17	17	14	11	9
		8 Supply	5	8	15	18	17	14	11	9
		6 Supply	6	9	16	19	17	14	11	9
		3 Supply	5	8	15	18	17	14	10	9
600ELF	600	0 Static	5	8	15	18	17	13	10	8
		3 Return	5	9	15	18	17	13	11	9
		6 Return	6	10	15	19	17	13	11	9
		8 Return	6	10	15	19	17	14	11	9
		10 Return	7	10	16	19	17	14	11	9
		10 Supply	7	11	20	26	26	22	16	13
		8 Supply	7	11	22	27	26	22	16	13
		6 Supply	7	12	24	28	26	22	16	13
		3 Supply	8	12	23	27	26	21	15	13
900ELF	900	0 Static	9	13	22	27	25	21	15	12
		3 Return	9	13	22	27	25	21	16	12
		6 Return	10	14	23	28	25	20	15	12
		8 Return	10	15	23	28	25	20	15	11
		10 Return	11	16	24	29	25	20	15	11
		10 Supply	9	15	24	29	32	25	18	15
		8 Supply	9	15	25	30	32	24	18	15
		6 Supply	9	16	27	32	33	24	18	15
		3 Supply	12	16	27	32	33	24	18	13
1200ELF	1200	0 Static	12	16	27	32	34	23	18	13
		3 Return	13	17	28	33	34	23	18	12
		6 Return	13	18	28	33	34	24	18	12
		8 Return	14	19	28	33	34	24	18	11
		10 Return	15	19	28	33	34	25	18	11
		10 Supply	11	19	27	32	37	28	21	17
		8 Supply	11	19	28	33	38	28	21	17
		6 Supply	11	20	30	35	40	28	21	17
		3 Supply	14	20	31	35	40	28	21	15
1500ELF	1500	0 Static	14	21	32	37	41	28	20	15
		3 Return	15	21	33	38	41	29	20	14
		6 Return	15	22	33	39	42	29	20	14
		8 Return	17	22	35	41	42	29	20	13
		10 Return	17	23	36	41	43	29	20	13

RECTANGULAR DUCT SILENCER TYPE
EXCELLENCE LOW-FREQUENCY NOISE ATTENUATION ELF

Model	Length (mm.)	Silencer Face Velocity m/s	63	125	250	500	1000	2000	4000	8000
			Octave Band Center Frequency, (Hz)							
		10 Supply	11	20	33	40	42	33	24	18
		8 Supply	11	21	34	40	42	33	24	18
		6 Supply	12	22	35	42	44	32	24	18
		3 Supply	14	22	36	42	44	32	23	18
1800ELF	1800	0 Static	15	23	38	42	44	31	23	17
		3 Return	15	24	38	43	46	31	24	17
		6 Return	15	24	38	44	46	32	24	17
		8 Return	17	24	39	46	46	32	24	16
		10 Return	17	24	40	46	47	32	25	16
		10 Supply	10	22	39	47	46	38	30	20
		8 Supply	10	23	40	47	46	38	30	20
		6 Supply	10	23	41	47	47	38	30	20
		3 Supply	12	25	41	47	47	36	27	19
2100ELF	2100	0 Static	12	25	43	47	47	36	26	19
		3 Return	15	26	43	48	51	36	25	18
		6 Return	15	26	43	48	52	36	25	18
		8 Return	17	26	43	50	52	36	24	17
		10 Return	18	26	43	50	53	36	23	17
		10 Supply	13	23	42	49	48	41	30	21
		8 Supply	13	24	43	49	48	41	30	21
		6 Supply	14	25	43	49	49	40	30	21
		3 Supply	15	26	43	49	49	39	29	20
2400ELF	2400	0 Static	16	28	45	49	49	39	27	19
		3 Return	17	28	45	49	51	39	28	18
		6 Return	17	28	45	49	52	39	29	17
		8 Return	18	28	45	51	52	39	28	17
		10 Return	18	28	45	51	53	39	28	17
		10 Supply	16	25	48	51	52	46	35	25
		8 Supply	17	27	48	51	52	46	35	24
		6 Supply	18	28	48	51	52	45	35	24
		3 Supply	19	28	48	52	53	45	34	21
3000ELF	3000	0 Static	21	29	48	52	54	45	32	20
		3 Return	21	32	49	52	54	44	30	18
		6 Return	21	32	49	53	54	44	29	18
		8 Return	21	33	50	53	55	43	27	18
		10 Return	21	34	51	54	55	42	27	16

Note :
Supply : Air is travelling in the same direction of sound propagation.
Return : Air is travelling in the opposite direction of sound propagation.
Interpolate dynamic insertion loss (DIL) from intermediate velocities.

RECTANGULAR DUCT SILENCER
EXCELLENCE LOW-FREQUENCY NOISE ATTENUATION — TYPE ELF

Abie Tiger

Duct Silencer Face Area (SQ.m.)	0.045	0.09	0.18	0.36	0.72	1.44	2.89	5.77	11.54
Self Noise Correction Factor, dB	-9	-6	-3	0	+3	+6	+9	+12	+15

Silencer Length (mm)	Static Pressure Drop, (Pa)											
600	5.9	13.2	23.5	36.8	52.9	94.1	147.0	211.7	288.1	376.3	476.3	588
900	6.1	13.7	24.3	38.0	54.7	97.2	151.9	218.7	297.7	388.9	492.2	607.6
1200	6.3	14.1	25.1	39.2	56.4	100.4	156.8	225.8	307.3	401.4	508.0	627.2
1500	6.5	14.6	25.9	40.4	58.2	103.5	161.7	232.8	316.9	414.0	523.9	646.8
1800	6.7	15.0	26.7	41.7	60.0	106.6	166.6	239.9	326.5	426.5	539.8	666.4
2100	6.9	15.4	27.4	42.9	61.7	109.8	171.5	247.0	336.1	439.0	555.7	686
2400	7.1	15.9	28.2	44.1	63.5	112.9	176.4	254.0	345.7	451.6	571.5	705.6
3000	7.3	16.3	29.0	45.3	65.3	116.0	181.3	261.1	355.3	464.1	587.4	725.2

Module Size width & Height (mm) x (mm)	Weight in kg per liner meter	Face Area sq.m.	Air Flow (cu.m./s.)											
300 x 300	20	0.09	0.09	0.14	0.18	0.23	0.27	0.36	0.45	0.54	0.63	0.72	0.81	0.90
300 x 450	30	0.14	0.14	0.20	0.27	0.34	0.41	0.54	0.68	0.81	0.95	1.08	1.22	1.35
300 x 600	40	0.18	0.18	0.27	0.36	0.45	0.54	0.72	0.90	1.08	1.26	1.44	1.62	1.80
300 x 750	50	0.23	0.23	0.34	0.45	0.56	0.68	0.90	1.13	1.35	1.58	1.80	2.03	2.25
300 x 900	60	0.27	0.27	0.41	0.54	0.68	0.81	1.08	1.35	1.62	1.89	2.16	2.43	2.70
300 x 1,050	70	0.32	0.32	0.47	0.63	0.79	0.95	1.26	1.58	1.89	2.21	2.52	2.84	3.15
300 x 1,200	80	0.36	0.36	0.54	0.72	0.90	1.08	1.44	1.80	2.16	2.52	2.88	3.24	3.60
600 x 450	60	0.27	0.27	0.41	0.54	0.68	0.81	1.08	1.35	1.62	1.89	2.16	2.43	2.70
600 x 600	80	0.36	0.36	0.54	0.72	0.92	1.08	1.44	1.80	2.16	2.52	2.88	3.24	3.60
600 x 750	100	0.45	0.45	0.68	0.90	1.13	1.35	1.80	2.25	2.70	3.15	3.60	4.05	4.50
600 x 900	120	0.54	0.54	0.81	1.08	1.35	1.62	2.16	2.70	3.24	3.78	4.32	4.86	5.40
600 x 1,050	140	0.63	0.63	0.95	1.26	1.58	1.89	2.52	3.15	3.78	4.41	4.04	5.67	6.30
600 x 1,200	160	0.72	0.72	1.08	1.44	1.80	2.16	2.88	3.60	4.32	5.04	5.76	6.48	7.20
1,200 x 750	200	0.9	0.9	1.35	1.80	2.25	2.70	3.60	4.50	5.40	6.30	7.20	8.10	9.00
1,200 x 900	240	1.08	1.08	1.62	2.16	2.70	3.24	4.32	5.40	6.48	7.56	8.64	9.72	10.80
1,200 x 1,050	280	1.26	1.26	1.89	2.52	3.15	3.78	5.04	6.30	7.56	8.82	10.08	11.34	12.60
1,200 x 1,200	320	1.44	1.44	2.16	2.88	3.60	4.32	5.76	7.20	8.64	10.08	11.52	12.96	14.40
Silencer entering Face Velocity, (m/s)			1	1.5	2	2.5	3	4	5	6	7	8	9	10

Note :

-If sound attenuators are installed immediately right after or before transitions, elbows, or at the intake or discharge of the system, the pressure drops must be changed from those given above by at least a factor at 1.5 times, depending upon specific conditions.

- Dynamic insertion loss and air flow data has been certified in accordance with BS 4718 and ASTM E477.

- All acoustic data has been determined in the Abie Tiger Acoustics Aero-Dynamic Reverberation Laboratory in a reverberant room of volume 332.6 m^3.

- All data was obtained on duct sizes 610 mm. x 610 mm.

HOW TO ORDER : TYPE EMF

EXAMPLE 1,200 X 1,000 EMF X 1,500
Duct Width x Duct Height x Model x Duct Length

Abie Tiger Acoustical Silencers

Silencer Splitters

Splitter is normally manufactured from minimum gauge 24 perforated galvanized steel with 2.6 mm hole diameter giving a minimum 17.23% open area. The bul nose in the machine side is fabricated from minimum gauge 24 galvanized steell, the bull nose at the other end is manufactured from same material as splitter. The splitters are inserted and locked into the duct silencer casing by an interlocking track. The track is from minimum gauge 24 solid galvanized steel. The infil material is absorption material tested in accordance with ASTM C423, ISO 354/1995 E in the reverberation chamber. The material is compressed to not less than 10% of its volume to ensure solid packing in the splitter.

Outer Casing

The outer casing is manufactured from minimum gauge 20 galvanized steel.

Silencer Performance

The acoustic silencer system is normally supplied as the module. Each module has been tested in according to ASTM E477-90a, BS 4718 in the reverberation laboratory for dynamic insertion loss, regenerated noise and air flow data. The silencer has been tested at the Warrington Fire Research Laboratories in Melbourne Australia and found to exceed the test requirements of BS 476 Part 24. This standard requires the silencer to be tested at a temperature of 650°C of 60 minutes and suffer no damage, distortion or buckling.

SELF-NOISE SOUND POWER LEVEL, dB 10^{-12} WATES

Model	Silencer	Octave Band Center Frequency, Hz							
	Face Velocity m/s	63	125	250	500	1000	2000	4000	8000
	10 Supply	52	49	46	42	41	44	40	40
	8 Supply	52	49	40	39	41	47	45	40
	6 Supply	43	37	35	33	36	31	25	21
EMF	3 Supply	28	20	20	<20	24	22	20	20
	3 Return	29	31	36	38	46	49	36	21
	6 Return	44	44	44	48	55	59	47	39
	8 Return	50	49	54	53	57	62	62	56
	10 Return	57	56	60	58	59	66	67	65

Supply : Air traveling in the same direction of sound propagation.
Return : Air traveling in the opposite direction of the sound propagation (Interpolate self-noise for intermediate velocities.)

DYNAMIC INSERTION LOSS (DIL) RATINGS. (dB)

Model	Length (mm.)	Silencer Face Velocity m/s	Octave Band Center Frequency, (Hz) 63	125	250	500	1000	2000	4000	8000
		10 Supply	3	5	9	17	21	23	15	11
		8 Supply	3	5	9	17	21	23	15	11
		6 Supply	3	5	9	17	23	22	15	11
		3 Supply	3	5	9	17	23	22	15	11
600EMF	600	0 Static	3	6	11	18	23	24	15	11
		3 Return	4	7	11	19	24	24	15	10
		6 Return	4	7	11	19	24	24	15	10
		8 Return	4	7	12	19	25	24	15	9
		10 Return	4	7	12	19	25	24	15	9
		10 Supply	3	6	14	22	30	32	23	17
		8 Supply	3	6	14	22	30	32	23	17
		6 Supply	3	7	15	24	32	32	23	16
		3 Supply	4	7	16	24	32	32	23	16
900EMF	900	0 Static	4	7	16	26	34	33	22	16
		3 Return	5	9	18	26	34	33	22	15
		6 Return	5	9	18	27	35	33	22	15
		8 Return	5	9	18	28	35	33	22	14
		10 Return	5	9	18	28	35	33	22	14
		10 Supply	6	10	17	29	40	41	30	22
		8 Supply	6	10	17	29	40	41	30	22
		6 Supply	6	10	17	30	41	41	29	21
		3 Supply	6	10	18	30	41	41	29	21
1200EMF	1200	0 Static	7	11	18	31	42	41	26	20
		3 Return	8	13	20	33	43	42	25	18
		6 Return	8	13	20	33	43	42	25	18
		8 Return	8	14	21	34	44	42	24	16
		10 Return	8	14	21	34	44	42	24	16
		10 Supply	8	11	18	32	47	47	35	26
		8 Supply	8	11	18	32	47	47	35	26
		6 Supply	8	12	19	34	47	47	34	25
		3 Supply	8	12	19	34	48	48	34	25
1500EMF	1500	0 Static	9	13	20	35	49	48	30	20
		3 Return	9	16	22	37	50	48	28	20
		6 Return	9	16	22	37	51	49	28	18
		8 Return	9	18	23	38	51	49	26	18
		10 Return	9	18	23	38	51	49	26	18

RECTANGULAR DUCT SILENCER TYPE
EXCELLENCE MID-FREQUENCY NOISE ATTENUATION EMF

Model	Length (mm.)	Silencer Face Velocity m/s	63	125	250	500	1000	2000	4000	8000
			Octave Band Center Frequency, (Hz)							
		10 Supply	8	13	25	40	50	48	41	29
		8 Supply	8	13	25	40	50	48	41	29
		6 Supply	8	14	20	42	51	48	40	29
		3 Supply	9	15	26	42	51	49	40	29
1800EMF	1800	0 Static	9	16	20	43	51	49	38	27
		3 Return	10	18	29	44	52	49	36	24
		6 Return	10	18	29	44	52	50	36	24
		8 Return	10	19	30	45	53	50	34	22
		10 Return	10	19	30	45	53	50	34	22
		10 Supply	9	15	31	47	53	50	46	32
		8 Supply	9	15	31	47	53	50	46	32
		6 Supply	9	15	32	49	53	51	45	30
		3 Supply	10	17	32	49	53	51	45	30
2100EMF	2100	0 Static	10	18	35	50	53	52	45	29
		3 Return	11	19	35	51	54	52	44	28
		6 Return	11	19	35	51	54	53	44	26
		8 Return	11	20	37	52	54	53	42	26
		10 Return	11	20	37	52	54	53	41	25
		10 Supply	10	18	34	50	54	54	48	35
		8 Supply	10	18	34	50	54	54	48	34
		6 Supply	10	18	35	51	54	54	47	34
		3 Supply	11	20	35	51	54	53	47	34
2400EMF	2400	0 Static	11	21	37	52	54	53	47	31
		3 Return	12	22	37	52	55	53	46	28
		6 Return	12	22	37	52	55	52	45	28
		8 Return	12	23	38	53	55	52	44	27
		10 Return	12	23	38	53	55	52	43	26
		10 Supply	9	24	38	53	53	56	53	40
		8 Supply	9	24	38	53	53	56	53	39
		6 Supply	9	25	38	53	53	55	52	39
		3 Supply	10	25	39	54	54	55	52	37
3000EMF	3000	0 Static	11	26	39	54	54	55	50	36
		3 Return	12	28	39	55	55	54	49	29
		6 Return	13	28	40	55	55	54	48	29
		8 Return	13	28	40	56	57	54	47	28
		10 Return	13	28	40	57	57	54	46	28

Note :
Supply : Air is travelling in the same direction of sound propagation.
Return : Air is travelling in the opposite direction of sound propagation.
Interpolate dynamic insertion loss (DIL) from intermediate velocities.

RECTANGULAR DUCT SILENCER
EXCELLENCE MID-FREQUENCY NOISE ATTENUATION TYPE EMF

Duct Silencer Face Area (SQ.m.)	0.045	0.09	0.18	0.36	0.72	1.44	2.89	5.77	11.54
Self Noise Correction Factor, dB	-9	-6	-3	0	+3	+6	+9	+12	+15

Silencer Length (mm)	Static Pressure Drop, (Pa)											
600	1.2	2.7	4.9	7.6	10.9	19.4	30.4	43.7	59.5	77.7	98.3	121.4
900	1.7	3.9	6.9	10.8	15.5	27.6	43.1	62.1	84.5	110.3	139.6	172.4
1200	2.2	5.0	8.9	14.0	20.1	35.7	55.9	80.4	109.5	143.0	181.0	223.4
1500	2.7	6.2	11.0	17.2	24.7	43.9	68.6	98.8	134.5	175.6	222.3	274.4
1800	3.3	7.3	13.0	20.3	29.3	52.1	81.4	117.1	159.4	208.3	263.6	325.4
2100	3.8	8.5	15.0	23.5	33.8	60.2	94.0	135.4	184.2	240.6	304.6	376
2400	4.3	9.6	17.1	26.8	38.5	68.5	107.0	154.1	209.7	273.9	346.7	428
3000	4.8	10.8	19.2	30.1	43.3	77.0	120.3	173.2	235.7	307.8	389.6	481

Module Size width & Height (mm) x (mm)	Weight in kg per liner meter	Face Area sq.m.	Air Flow (cu.m./s.)											
300 x 300	20	0.09	0.09	0.14	0.18	0.23	0.27	0.36	0.45	0.54	0.63	0.72	0.81	0.90
300 x 450	30	0.14	0.14	0.20	0.27	0.34	0.41	0.54	0.68	0.81	0.95	1.08	1.22	1.35
300 x 600	40	0.18	0.18	0.27	0.36	0.45	0.54	0.72	0.90	1.08	1.26	1.44	1.62	1.80
300 x 750	50	0.23	0.23	0.34	0.45	0.56	0.68	0.90	1.13	1.35	1.58	1.80	2.03	2.25
300 x 900	60	0.27	0.27	0.41	0.54	0.68	0.81	1.08	1.35	1.62	1.89	2.16	2.43	2.70
300 x 1,050	70	0.32	0.32	0.47	0.63	0.79	0.95	1.26	1.58	1.89	2.21	2.52	2.84	3.15
300 x 1,200	80	0.36	0.36	0.54	0.72	0.90	1.08	1.44	1.80	2.16	2.52	2.88	3.24	3.60
600 x 450	60	0.27	0.27	0.41	0.54	0.68	0.81	1.08	1.35	1.62	1.89	2.16	2.43	2.70
600 x 600	80	0.36	0.36	0.54	0.72	0.92	1.08	1.44	1.80	2.16	2.52	2.88	3.24	3.60
600 x 750	100	0.45	0.45	0.68	0.90	1.13	1.35	1.80	2.25	2.70	3.15	3.60	4.05	4.50
600 x 900	120	0.54	0.54	0.81	1.08	1.35	1.62	2.16	2.70	3.24	3.78	4.32	4.86	5.40
600 x 1,050	140	0.63	0.63	0.95	1.26	1.58	1.89	2.52	3.15	3.78	4.41	4.04	5.67	6.30
600 x 1,200	160	0.72	0.72	1.08	1.44	1.80	2.16	2.88	3.60	4.32	5.04	5.76	6.48	7.20
1,200 x 750	200	0.9	0.9	1.35	1.80	2.25	2.70	3.60	4.50	5.40	6.30	7.20	8.10	9.00
1,200 x 900	240	1.08	1.08	1.62	2.16	2.70	3.24	4.32	5.40	6.48	7.56	8.64	9.72	10.80
1,200 x 1,050	280	1.26	1.26	1.89	2.52	3.15	3.78	5.04	6.30	7.56	8.82	10.08	11.34	12.60
1,200 x 1,200	320	1.44	1.44	2.16	2.88	3.60	4.32	5.76	7.20	8.64	10.08	11.52	12.96	14.40
Silencer entering Face Velocity, (m/s)			1	1.5	2	2.5	3	4	5	6	7	8	9	10

Note :

-If sound attenuators are installed immediately right after or before transitions, elbows, or at the intake or discharge of the system, the pressure drops must be changed from those given above by at least a factor at 1.5 times, depending upon specific conditions.

- Dynamic insertion loss and air flow data has been certified in accordance with BS 4718 and ASTM E477.

- All acoustic data has been determined in the Abie Tiger Acoustics Aero-Dynamic Reverberation Laboratory in a reverberant room of volume 332.6 m^3.

- All data was obtained on duct sizes 610 mm. x 610 mm.

HOW TO ORDER : TYPE SF

EXAMPLE 1,200 X 1,000 EMF X 1,500
Duct Width x Duct Height x Model x Duct Length

Abie Tiger Acoustical Silencers

Silencer Splitters

Splitter is normally manufactured from minimum gauge 24 perforated galvanized steel with 2.6 mm hole diameter giving a minimum 17.23% open area. The bul nose in the machine side is fabricated from minimum gauge 24 galvanized steell, the bull nose at the other end is manufactured from same material as splitter. The splitters are inserted and locked into the duct silencer casing by an interlocking track. The track is from minimum gauge 24 solid galvanized steel. The infil material is absorption material tested in accordance with ASTM C423, ISO 354/1995 E in the reverberation chamber. The material is compressed to not less than 10% of its volume to ensure solid packing in the splitter.

Outer Casing

The outer casing is manufactured from minimum gauge 20 galvanized steel.

Silencer Performance

The acoustic silencer system is normally supplied as the module. Each module has been tested in according to ASTM E477-90a, BS 4718 in the reverberation laboratory for dynamic insertion loss, regenerated noise and air flow data. The silencer has been tested at the Warrington Fire Research Laboratories in Melbourne Australia and found to exceed the test requirements of BS 476 Part 24. This standard requires the silencer to be tested at a temperature of 650°C of 60 minutes and suffer no damage, distortion or buckling.

SELF-NOISE SOUND POWER LEVEL, dB 10^{-12} WATES

Model	Silencer	Octave Band Center Frequency, Hz							
	Face Velocity m/s	63	125	250	500	1000	2000	4000	8000
	10 Supply	72	66	63	66	62	65	64	57
	8 Supply	68	63	58	59	56	58	55	48
	6 Supply	57	51	51	49	47	51	44	33
SF	3 Supply	46	42	40	41	41	42	35	23
	3 Return	46	44	43	45	50	51	47	31
	6 Return	52	53	48	52	52	54	54	42
	8 Return	64	59	58	60	60	62	62	57
	10 Return	69	63	59	64	59	66	66	64

Supply : Air traveling in the same direction of sound propagatin.
Return : Air traveling in the opposite direction of the sound propagation (Interpolate self-noise for intermediate velocities.)

DYNAMIC INSERTION LOSS (DIL) RATINGS. (dB)

Model	Length (mm.)	Silencer Face Velocity m/s	63	125	250	500	1000	2000	4000	8000
		10 Supply	5	6	11	17	23	23	19	11
		8 Supply	5	6	11	17	23	23	19	11
		6 Supply	5	7	11	17	23	23	19	11
		3 Supply	5	7	11	17	23	23	18	11
600SF	600	0 Static	5	7	11	18	24	22	18	10
		3 Return	5	7	12	20	24	21	17	9
		6 Return	5	8	13	20	24	21	17	9
		8 Return	5	8	13	20	24	21	17	9
		10 Return	5	9	14	20	24	21	17	9
		10 Supply	7	9	14	23	32	34	29	16
		8 Supply	7	9	14	23	32	34	28	16
		6 Supply	7	10	15	23	33	34	27	16
		3 Supply	7	10	15	24	33	34	27	16
900SF	900	0 Static	6	11	17	27	34	33	27	15
		3 Return	7	11	18	28	35	31	25	14
		6 Return	7	12	19	28	35	31	25	14
		8 Return	7	12	20	30	36	31	25	12
		10 Return	7	13	20	30	36	31	25	11
		10 Supply	7	13	19	31	40	41	37	20
		8 Supply	7	13	19	31	40	41	36	20
		6 Supply	7	14	19	31	40	41	36	20
		3 Supply	7	14	19	32	40	41	36	20
1200SF	1200	0 Static	8	15	20	33	41	39	36	19
		3 Return	9	17	22	36	41	37	34	18
		6 Return	9	17	22	37	41	37	34	18
		8 Return	9	18	22	37	41	37	34	18
		10 Return	9	18	23	37	41	37	34	18
		10 Supply	6	17	21	37	44	44	40	24
		8 Supply	6	17	21	37	44	44	40	24
		6 Supply	6	17	22	37	44	44	40	24
		3 Supply	6	17	22	38	44	43	39	23
1500SF	1500	0 Static	7	18	23	38	45	43	39	23
		3 Return	10	21	24	41	46	43	38	22
		6 Return	10	21	24	41	46	42	38	22
		8 Return	10	21	25	42	46	42	36	20
		10 Return	10	21	25	42	46	42	35	20

Model	Length (mm.)	Silencer Face Velocity m/s	63	125	250	500	1000	2000	4000	8000
			Octave Band Center Frequency, (Hz)							
		10 Supply	7	18	27	42	47	46	42	30
		8 Supply	7	18	27	42	47	46	42	30
		6 Supply	7	18	28	42	47	46	42	30
		3 Supply	8	18	28	43	47	45	41	28
1800SF	1800	0 Static	8	20	29	43	48	45	41	28
		3 Return	9	22	31	43	50	44	41	26
		6 Return	9	22	31	44	50	44	40	26
		8 Return	10	22	32	44	50	44	40	25
		10 Return	10	22	32	44	50	44	40	25
		10 Supply	8	19	33	44	50	48	44	33
		8 Supply	8	19	33	44	50	48	44	33
		6 Supply	9	19	33	44	50	48	44	33
		3 Supply	9	21	33	44	50	48	43	32
2100SF	2100	0 Static	10	23	34	46	50	47	42	32
		3 Return	10	23	36	46	51	47	42	32
		6 Return	11	23	36	46	51	47	42	31
		8 Return	11	23	37	46	51	47	42	31
		10 Return	11	23	37	46	52	47	42	31
		10 Supply	11	20	36	46	51	50	47	39
		8 Supply	11	20	36	46	51	50	47	39
		6 Supply	11	20	36	46	51	49	46	37
		3 Supply	11	22	36	46	51	49	46	37
2400SF	2400	0 Static	12	24	37	47	52	49	45	35
		3 Return	12	24	38	48	54	49	44	34
		6 Return	12	24	38	49	54	49	44	34
		8 Return	12	24	39	49	54	49	44	34
		10 Return	12	24	39	50	54	49	44	34
		10 Supply	11	21	39	51	53	54	50	47
		8 Supply	11	21	39	51	53	54	50	47
		6 Supply	11	21	39	51	53	54	50	44
		3 Supply	12	23	40	51	53	54	50	44
3000SF	3000	0 Static	12	25	40	52	54	52	50	43
		3 Return	12	25	41	53	54	52	49	41
		6 Return	12	25	41	53	56	52	49	41
		8 Return	12	25	41	53	56	52	49	41
		10 Return	12	25	42	53	56	52	48	41

Note :
Supply : Air is travelling in the same direction of sound propagation.
Return : Air is travelling in the opposite direction of sound propagation.
Interpolate dynamic insertion loss (DIL) from intermediate velocities.

RECTANGULAR DUCT SILENCER
SUPPREME FREQUENCY NOISE ATTENUATION

TYPE **SF**

Duct Silencer Face Area (SQ.m.)	0.045	0.09	0.18	0.36	0.72	1.44	2.89	5.77	11.54
Self Noise Correction Factor, dB	-9	-6	-3	0	+3	+6	+9	+12	+15

Silencer Length (mm)	Static Pressure Drop, (Pa)											
600	3.7	8.2	14.6	22.9	32.9	58.6	91.5	131.8	179.3	234.2	296.5	366
900	3.9	8.8	15.7	24.5	35.3	62.7	100.0	141.1	192.1	250.9	317.5	392
1200	4.2	9.4	16.7	26.1	37.6	66.9	104.5	150.5	204.8	267.5	338.6	418
1500	4.4	10.0	17.8	27.8	40.0	71.0	111.0	159.8	217.6	284.2	359.6	444
1800	4.7	10.6	18.8	29.4	42.3	75.2	117.5	169.2	230.3	300.8	380.7	470
2100	5.0	11.2	19.9	31.1	44.7	79.5	124.3	178.9	243.5	318.1	402.6	497
2400	5.2	11.8	20.9	32.7	47.1	83.7	130.8	188.3	256.3	334.7	423.6	523
3000	5.5	12.4	22.0	34.3	49.4	87.8	137.3	197.6	269.0	351.4	444.7	549

Module Size width & Height (mm) x (mm)	Weight in kg per liner meter	Face Area sq.m.	Air Flow (cu.m./s.)											
300 x 300	20	0.09	0.09	0.14	0.18	0.23	0.27	0.36	0.45	0.54	0.63	0.72	0.81	0.90
300 x 450	30	0.14	0.14	0.20	0.27	0.34	0.41	0.54	0.68	0.81	0.95	1.08	1.22	1.35
300 x 600	40	0.18	0.18	0.27	0.36	0.45	0.54	0.72	0.90	1.08	1.26	1.44	1.62	1.80
300 x 750	50	0.23	0.23	0.34	0.45	0.56	0.68	0.90	1.13	1.35	1.58	1.80	2.03	2.25
300 x 900	60	0.27	0.27	0.41	0.54	0.68	0.81	1.08	1.35	1.62	1.89	2.16	2.43	2.70
300 x 1,050	70	0.32	0.32	0.47	0.63	0.79	0.95	1.26	1.58	1.89	2.21	2.52	2.84	3.15
300 x 1,200	80	0.36	0.36	0.54	0.72	0.90	1.08	1.44	1.80	2.16	2.52	2.88	3.24	3.60
600 x 450	60	0.27	0.27	0.41	0.54	0.68	0.81	1.08	1.35	1.62	1.89	2.16	2.43	2.70
600 x 600	80	0.36	0.36	0.54	0.72	0.92	1.08	1.44	1.80	2.16	2.52	2.88	3.24	3.60
600 x 750	100	0.45	0.45	0.68	0.90	1.13	1.35	1.80	2.25	2.70	3.15	3.60	4.05	4.50
600 x 900	120	0.54	0.54	0.81	1.08	1.35	1.62	2.16	2.70	3.24	3.78	4.32	4.86	5.40
600 x 1,050	140	0.63	0.63	0.95	1.26	1.58	1.89	2.52	3.15	3.78	4.41	4.04	5.67	6.30
600 x 1,200	160	0.72	0.72	1.08	1.44	1.80	2.16	2.88	3.60	4.32	5.04	5.76	6.48	7.20
1,200 x 750	200	0.9	0.9	1.35	1.80	2.25	2.70	3.60	4.50	5.40	6.30	7.20	8.10	9.00
1,200 x 900	240	1.08	1.08	1.62	2.16	2.70	3.24	4.32	5.40	6.48	7.56	8.64	9.72	10.80
1,200 x 1,050	280	1.26	1.26	1.89	2.52	3.15	3.78	5.04	6.30	7.56	8.82	10.08	11.34	12.60
1,200 x 1,200	320	1.44	1.44	2.16	2.88	3.60	4.32	5.76	7.20	8.64	10.08	11.52	12.96	14.40
Silencer entering Face Velocity, (m/s)			1	1.5	2	2.5	3	4	5	6	7	8	9	10

Note :

-If sound attenuators are installed immediately right after or before transitions, elbows, or at the intake or discharge of the system, the pressure drops must be changed from those given above by at least a factor at 1.5 times, depending upon specific conditions.

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WB1

ABIE TIGER PRODUCTS:-


Performance Doors

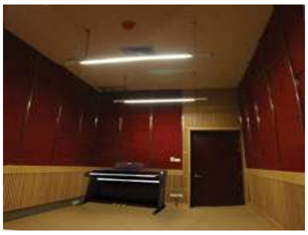

Acoustic Ceilings


Acoustic Barrier wall Panels


Acoustic Wall Panels


Acoustic Machine Enclousures


Engine Test Cell Enclosures


Music Practise Enclosures

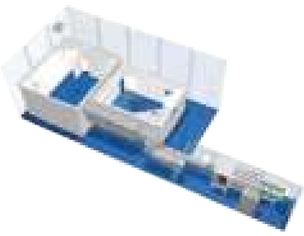

Audiometric Booths

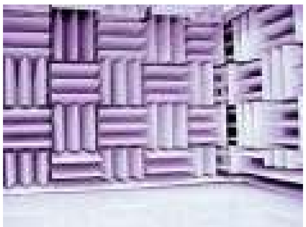

Acoustic Metal Composite Roofs


Fire Acoustic Access Floors


Duct Silencers


Acoustic Glazing


Reverberation Chambers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Noise Blocks


Mufflers


Acoustic Louvers


Performance Air Grills


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